Um breve tutorial sobre dplyr com ggplot2

true

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Chapter 1

Introdução

Palestra do *XIV Programa de Verão DES-ICET/UFLA - 2023*

1.1 Alguns atalhos no Rstudio

Para considerar

> Operador Pipe (%>%): Ctrl + Shift + M (Windows) ou Cmd + Shift + M (Mac).
>
> Criar novos chunks: Cmd + Option + I (Windows Ctrl + Alt + I).

1.2 Carrega pacotes a serem usados

```
#install.packages("tidyverse")
#install.packages("dplyr")
#install.packages("tidyr")
#install.packages("ggplot2")

# Manipulação de dados
library(dplyr)

# Visualização de gráficos
library(ggplot2)
library(gridExtra)
library(patchwork)
library(plotly)
library(esquisse)
```

1.3 Descrição dos dados `mpg`

Fuel economy data from 1999 to 2008 for 38 popular models of cars. This dataset contains a subset of the fuel economy data that the EPA makes available on https://fueleconomy.gov/. It contains only models which had a new release every year between 1999 and 2008 - this was used as a proxy for the popularity of the car. A data frame with 234 rows and 11 variables:

- manufacturer manufacturer name
- model model name
- displ engine displacement, in litres
- year year of manufacture
- cyl number of cylinders
- trans type of transmission
- drv the type of drive train, where f = front-wheel drive, r = rear wheel drive, 4 = 4wd
- cty city miles per gallon
- hwy highway miles per gallon
- fl fuel type
- class "type" of car

```
#help("mpg")
dados <- mpg
#View(df)
glimpse(dados)
```

```
## Rows: 234
## Columns: 11
## $ manufacturer <chr> "audi", "audi", "audi", "audi", "audi", "audi", "audi", "~
## $ model        <chr> "a4", "a4", "a4", "a4", "a4", "a4", "a4", "a4 quattro", "~
## $ displ        <dbl> 1.8, 1.8, 2.0, 2.0, 2.8, 2.8, 3.1, 1.8, 1.8, 2.0, 2.0, 2.~
## $ year         <int> 1999, 1999, 2008, 2008, 1999, 1999, 2008, 1999, 1999, 200~
## $ cyl          <int> 4, 4, 4, 4, 6, 6, 6, 4, 4, 4, 4, 6, 6, 6, 6, 6, 6, 8, 8, ~
## $ trans        <chr> "auto(l5)", "manual(m5)", "manual(m6)", "auto(av)", "auto~
## $ drv          <chr> "f", "f", "f", "f", "f", "f", "f", "4", "4", "4", "4", "4~
## $ cty          <int> 18, 21, 20, 21, 16, 18, 18, 18, 16, 20, 19, 15, 17, 17, 1~
## $ hwy          <int> 29, 29, 31, 30, 26, 26, 27, 26, 25, 28, 27, 25, 25, 25, 2~
## $ fl           <chr> "p", "p", "p", "p", "p", "p", "p", "p", "p", "p", "p", "p~
## $ class        <chr> "compact", "compact", "compact", "compact", "compact", "c~
```

1.4 Propaganda 1 (Gustavo Jun Yakushiji)

Assistir os 3 vídeos do **Gustavo Jun Yakushiji** e **Cristian Villegas** no youtube

Dia 1 Introdução à Ciência de Dados (Introdução ao R e importação de dados)

Dia 2 Introdução à Ciência de Dados (Introdução ao ggplot2)

Dia 3 Introdução à Ciência de Dados (Introdução ao Tidyverse)

1.5 Propaganda 2 (Gustavo Jun Yakushiji)

Introdução à Ciência de Dados em R, Gustavo Jun Yakushiji; Cristian Villegas

Chapter 2

dplyr (60 minutos)

```r
ls("package:dplyr")
```

```
##   [1] "%>%"                   "across"                "add_count"
##   [4] "add_count_"            "add_row"               "add_rownames"
##   [7] "add_tally"             "add_tally_"            "all_equal"
##  [10] "all_of"                "all_vars"              "anti_join"
##  [13] "any_of"                "any_vars"              "arrange"
##  [16] "arrange_"              "arrange_all"           "arrange_at"
##  [19] "arrange_if"            "as.tbl"                "as_data_frame"
##  [22] "as_label"              "as_tibble"             "auto_copy"
##  [25] "band_instruments"      "band_instruments2"     "band_members"
##  [28] "bench_tbls"            "between"               "bind_cols"
##  [31] "bind_rows"             "c_across"              "case_when"
##  [34] "changes"               "check_dbplyr"          "coalesce"
##  [37] "collapse"              "collect"               "combine"
##  [40] "common_by"             "compare_tbls"          "compare_tbls2"
##  [43] "compute"               "contains"              "copy_to"
##  [46] "count"                 "count_"                "cumall"
##  [49] "cumany"                "cume_dist"             "cummean"
##  [52] "cur_column"            "cur_data"              "cur_data_all"
##  [55] "cur_group"             "cur_group_id"          "cur_group_rows"
##  [58] "current_vars"          "data_frame"            "data_frame_"
##  [61] "db_analyze"            "db_begin"              "db_commit"
##  [64] "db_create_index"       "db_create_indexes"     "db_create_table"
##  [67] "db_data_type"          "db_desc"               "db_drop_table"
##  [70] "db_explain"            "db_has_table"          "db_insert_into"
##  [73] "db_list_tables"        "db_query_fields"       "db_query_rows"
##  [76] "db_rollback"           "db_save_query"         "db_write_table"
##  [79] "dense_rank"            "desc"                  "dim_desc"
```

```
##  [82] "distinct"              "distinct_"             "distinct_all"
##  [85] "distinct_at"           "distinct_if"           "distinct_prepare"
##  [88] "do"                    "do_"                   "dplyr_col_modify"
##  [91] "dplyr_reconstruct"     "dplyr_row_slice"       "ends_with"
##  [94] "enexpr"                "enexprs"               "enquo"
##  [97] "enquos"                "ensym"                 "ensyms"
## [100] "eval_tbls"             "eval_tbls2"            "everything"
## [103] "explain"               "expr"                  "failwith"
## [106] "filter"                "filter_"               "filter_all"
## [109] "filter_at"             "filter_if"             "first"
## [112] "frame_data"            "full_join"             "funs"
## [115] "funs_"                 "glimpse"               "group_by"
## [118] "group_by_"             "group_by_all"          "group_by_at"
## [121] "group_by_drop_default" "group_by_if"           "group_by_prepare"
## [124] "group_cols"            "group_data"            "group_indices"
## [127] "group_indices_"        "group_keys"            "group_map"
## [130] "group_modify"          "group_nest"            "group_rows"
## [133] "group_size"            "group_split"           "group_trim"
## [136] "group_vars"            "group_walk"            "grouped_df"
## [139] "groups"                "id"                    "ident"
## [142] "if_all"                "if_any"                "if_else"
## [145] "inner_join"            "intersect"             "is.grouped_df"
## [148] "is.src"                "is.tbl"                "is_grouped_df"
## [151] "lag"                   "last"                  "last_col"
## [154] "lead"                  "left_join"             "location"
## [157] "lst"                   "lst_"                  "make_tbl"
## [160] "matches"               "min_rank"              "mutate"
## [163] "mutate_"               "mutate_all"            "mutate_at"
## [166] "mutate_each"           "mutate_each_"          "mutate_if"
## [169] "n"                     "n_distinct"            "n_groups"
## [172] "na_if"                 "near"                  "nest_by"
## [175] "nest_join"             "new_grouped_df"        "new_rowwise_df"
## [178] "nth"                   "ntile"                 "num_range"
## [181] "one_of"                "order_by"              "percent_rank"
## [184] "progress_estimated"    "pull"                  "quo"
## [187] "quo_name"              "quos"                  "recode"
## [190] "recode_factor"         "relocate"              "rename"
## [193] "rename_"               "rename_all"            "rename_at"
## [196] "rename_if"             "rename_vars"           "rename_vars_"
## [199] "rename_with"           "right_join"            "row_number"
## [202] "rows_append"           "rows_delete"           "rows_insert"
## [205] "rows_patch"            "rows_update"           "rows_upsert"
## [208] "rowwise"               "same_src"              "sample_frac"
## [211] "sample_n"              "select"                "select_"
## [214] "select_all"            "select_at"             "select_if"
## [217] "select_var"            "select_vars"           "select_vars_"
```

```
## [220] "semi_join"            "setdiff"               "setequal"
## [223] "show_query"           "slice"                 "slice_"
## [226] "slice_head"           "slice_max"             "slice_min"
## [229] "slice_sample"         "slice_tail"            "sql"
## [232] "sql_escape_ident"     "sql_escape_string"     "sql_join"
## [235] "sql_select"           "sql_semi_join"         "sql_set_op"
## [238] "sql_subquery"         "sql_translate_env"     "src"
## [241] "src_df"               "src_local"             "src_mysql"
## [244] "src_postgres"         "src_sqlite"            "src_tbls"
## [247] "starts_with"          "starwars"              "storms"
## [250] "summarise"            "summarise_"            "summarise_all"
## [253] "summarise_at"         "summarise_each"        "summarise_each_"
## [256] "summarise_if"         "summarize"             "summarize_"
## [259] "summarize_all"        "summarize_at"          "summarize_each"
## [262] "summarize_each_"      "summarize_if"          "sym"
## [265] "syms"                 "tally"                 "tally_"
## [268] "tbl"                  "tbl_df"                "tbl_nongroup_vars"
## [271] "tbl_ptype"            "tbl_sum"               "tbl_vars"
## [274] "tibble"               "top_frac"              "top_n"
## [277] "transmute"            "transmute_"            "transmute_all"
## [280] "transmute_at"         "transmute_if"          "tribble"
## [283] "type_sum"             "ungroup"               "union"
## [286] "union_all"            "validate_grouped_df"   "validate_rowwise_df"
## [289] "vars"                 "with_groups"           "with_order"
## [292] "wrap_dbplyr_obj"
```

2.1 Operador Pipe

```r
sqrt(log(44))
```

```
## [1] 1.945299
```

```r
44 %>% log %>% sqrt
```

```
## [1] 1.945299
```

2.2 select() para linhas

```r
select(dados, manufacturer, model, year)
```

```
## # A tibble: 234 x 3
##    manufacturer model       year
##    <chr>        <chr>      <int>
##  1 audi         a4          1999
##  2 audi         a4          1999
```

```
##  3 audi         a4          2008
##  4 audi         a4          2008
##  5 audi         a4          1999
##  6 audi         a4          1999
##  7 audi         a4          2008
##  8 audi         a4 quattro  1999
##  9 audi         a4 quattro  1999
## 10 audi         a4 quattro  2008
## # ... with 224 more rows
```

```
select(dados, starts_with("m"))
```

```
## # A tibble: 234 x 2
##    manufacturer model
##    <chr>        <chr>
##  1 audi         a4
##  2 audi         a4
##  3 audi         a4
##  4 audi         a4
##  5 audi         a4
##  6 audi         a4
##  7 audi         a4
##  8 audi         a4 quattro
##  9 audi         a4 quattro
## 10 audi         a4 quattro
## # ... with 224 more rows
```

```
select(dados, contains("r"))
```

```
## # A tibble: 234 x 4
##    manufacturer  year trans      drv
##    <chr>        <int> <chr>      <chr>
##  1 audi          1999 auto(l5)   f
##  2 audi          1999 manual(m5) f
##  3 audi          2008 manual(m6) f
##  4 audi          2008 auto(av)   f
##  5 audi          1999 auto(l5)   f
##  6 audi          1999 manual(m5) f
##  7 audi          2008 auto(av)   f
##  8 audi          1999 manual(m5) 4
##  9 audi          1999 auto(l5)   4
## 10 audi          2008 manual(m6) 4
## # ... with 224 more rows
```

```
select(dados, ends_with("y"))
```

```
## # A tibble: 234 x 2
##      cty   hwy
```

```
##    <int> <int>
##  1    18    29
##  2    21    29
##  3    20    31
##  4    21    30
##  5    16    26
##  6    18    26
##  7    18    27
##  8    18    26
##  9    16    25
## 10    20    28
## # ... with 224 more rows
```

```
select(dados, matches("[abc]"))
```

```
## # A tibble: 234 x 6
##    manufacturer  year   cyl trans        cty class
##    <chr>        <int> <int> <chr>      <int> <chr>
##  1 audi          1999     4 auto(l5)      18 compact
##  2 audi          1999     4 manual(m5)    21 compact
##  3 audi          2008     4 manual(m6)    20 compact
##  4 audi          2008     4 auto(av)      21 compact
##  5 audi          1999     6 auto(l5)      16 compact
##  6 audi          1999     6 manual(m5)    18 compact
##  7 audi          2008     6 auto(av)      18 compact
##  8 audi          1999     4 manual(m5)    18 compact
##  9 audi          1999     4 auto(l5)      16 compact
## 10 audi          2008     4 manual(m6)    20 compact
## # ... with 224 more rows
```

```
select(dados, starts_with("m"), starts_with("c"))
```

```
## # A tibble: 234 x 5
##    manufacturer model        cyl   cty class
##    <chr>        <chr>      <int> <int> <chr>
##  1 audi         a4             4    18 compact
##  2 audi         a4             4    21 compact
##  3 audi         a4             4    20 compact
##  4 audi         a4             4    21 compact
##  5 audi         a4             6    16 compact
##  6 audi         a4             6    18 compact
##  7 audi         a4             6    18 compact
##  8 audi         a4 quattro     4    18 compact
##  9 audi         a4 quattro     4    16 compact
## 10 audi         a4 quattro     4    20 compact
## # ... with 224 more rows
```

```r
select(dados, ends_with("l"), ends_with("s"))
```

```
## # A tibble: 234 x 6
##    model      displ   cyl fl    trans      class
##    <chr>      <dbl> <int> <chr> <chr>      <chr>
##  1 a4           1.8     4 p     auto(l5)   compact
##  2 a4           1.8     4 p     manual(m5) compact
##  3 a4           2       4 p     manual(m6) compact
##  4 a4           2       4 p     auto(av)   compact
##  5 a4           2.8     6 p     auto(l5)   compact
##  6 a4           2.8     6 p     manual(m5) compact
##  7 a4           3.1     6 p     auto(av)   compact
##  8 a4 quattro   1.8     4 p     manual(m5) compact
##  9 a4 quattro   1.8     4 p     auto(l5)   compact
## 10 a4 quattro   2       4 p     manual(m6) compact
## # ... with 224 more rows
```

```r
select(dados, 1:3)
```

```
## # A tibble: 234 x 3
##    manufacturer model      displ
##    <chr>        <chr>      <dbl>
##  1 audi         a4           1.8
##  2 audi         a4           1.8
##  3 audi         a4           2
##  4 audi         a4           2
##  5 audi         a4           2.8
##  6 audi         a4           2.8
##  7 audi         a4           3.1
##  8 audi         a4 quattro   1.8
##  9 audi         a4 quattro   1.8
## 10 audi         a4 quattro   2
## # ... with 224 more rows
```

```r
select(dados, c(2,5,7))
```

```
## # A tibble: 234 x 3
##    model        cyl drv
##    <chr>      <int> <chr>
##  1 a4             4 f
##  2 a4             4 f
##  3 a4             4 f
##  4 a4             4 f
##  5 a4             6 f
##  6 a4             6 f
##  7 a4             6 f
##  8 a4 quattro     4 4
```

```
##  9 a4 quattro     4 4
## 10 a4 quattro     4 4
## # ... with 224 more rows
```

```
select(dados, manufacturer:cyl)
```

```
## # A tibble: 234 x 5
##    manufacturer model      displ  year   cyl
##    <chr>        <chr>      <dbl> <int> <int>
##  1 audi         a4           1.8  1999     4
##  2 audi         a4           1.8  1999     4
##  3 audi         a4           2    2008     4
##  4 audi         a4           2    2008     4
##  5 audi         a4           2.8  1999     6
##  6 audi         a4           2.8  1999     6
##  7 audi         a4           3.1  2008     6
##  8 audi         a4 quattro   1.8  1999     4
##  9 audi         a4 quattro   1.8  1999     4
## 10 audi         a4 quattro   2    2008     4
## # ... with 224 more rows
```

```
select(dados,-(manufacturer:cyl))
```

```
## # A tibble: 234 x 6
##    trans      drv     cty   hwy fl    class
##    <chr>      <chr> <int> <int> <chr> <chr>
##  1 auto(l5)   f        18    29 p     compact
##  2 manual(m5) f        21    29 p     compact
##  3 manual(m6) f        20    31 p     compact
##  4 auto(av)   f        21    30 p     compact
##  5 auto(l5)   f        16    26 p     compact
##  6 manual(m5) f        18    26 p     compact
##  7 auto(av)   f        18    27 p     compact
##  8 manual(m5) 4        18    26 p     compact
##  9 auto(l5)   4        16    25 p     compact
## 10 manual(m6) 4        20    28 p     compact
## # ... with 224 more rows
```

2.3 rename()

```
dados1 <- rename(dados,
              mnfc = manufacturer,
              mod = model)
dados1
```

```
## # A tibble: 234 x 11
```

```
##    mnfc  mod        displ  year   cyl trans      drv     cty   hwy fl    class
##    <chr> <chr>      <dbl> <int> <int> <chr>      <chr> <int> <int> <chr> <chr>
##  1 audi  a4           1.8  1999     4 auto(l5)   f        18    29 p     compact
##  2 audi  a4           1.8  1999     4 manual(m5) f        21    29 p     compact
##  3 audi  a4           2    2008     4 manual(m6) f        20    31 p     compact
##  4 audi  a4           2    2008     4 auto(av)   f        21    30 p     compact
##  5 audi  a4           2.8  1999     6 auto(l5)   f        16    26 p     compact
##  6 audi  a4           2.8  1999     6 manual(m5) f        18    26 p     compact
##  7 audi  a4           3.1  2008     6 auto(av)   f        18    27 p     compact
##  8 audi  a4 quattro   1.8  1999     4 manual(m5) 4        18    26 p     compact
##  9 audi  a4 quattro   1.8  1999     4 auto(l5)   4        16    25 p     compact
## 10 audi  a4 quattro   2    2008     4 manual(m6) 4        20    28 p     compact
## # ... with 224 more rows
```

```
select(dados,
       mnfc = manufacturer,
       mod = model)
```

```
## # A tibble: 234 x 2
##    mnfc  mod
##    <chr> <chr>
##  1 audi  a4
##  2 audi  a4
##  3 audi  a4
##  4 audi  a4
##  5 audi  a4
##  6 audi  a4
##  7 audi  a4
##  8 audi  a4 quattro
##  9 audi  a4 quattro
## 10 audi  a4 quattro
## # ... with 224 more rows
```

```
select(dados,
       mnfc = manufacturer,
       mod = model,
       everything())
```

```
## # A tibble: 234 x 11
##    mnfc  mod        displ  year   cyl trans      drv     cty   hwy fl    class
##    <chr> <chr>      <dbl> <int> <int> <chr>      <chr> <int> <int> <chr> <chr>
##  1 audi  a4           1.8  1999     4 auto(l5)   f        18    29 p     compact
##  2 audi  a4           1.8  1999     4 manual(m5) f        21    29 p     compact
##  3 audi  a4           2    2008     4 manual(m6) f        20    31 p     compact
##  4 audi  a4           2    2008     4 auto(av)   f        21    30 p     compact
##  5 audi  a4           2.8  1999     6 auto(l5)   f        16    26 p     compact
##  6 audi  a4           2.8  1999     6 manual(m5) f        18    26 p     compact
```

```
##  7 audi  a4           3.1  2008     6 auto(av)   f        18    27 p     compact
##  8 audi  a4 quattro   1.8  1999     4 manual(m5) 4        18    26 p     compact
##  9 audi  a4 quattro   1.8  1999     4 auto(l5)   4        16    25 p     compact
## 10 audi  a4 quattro   2    2008     4 manual(m6) 4        20    28 p     compact
## # ... with 224 more rows
```

2.4 mutate() para colunas

```
dados <- mutate(dados,
             `avg miles per gallon` = (cty + hwy) / 2)
dados
```

```
## # A tibble: 234 x 12
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 224 more rows, and abbreviated variable names 1: manufacturer,
## #   2: `avg miles per gallon`
```

```
dados <- mutate(dados,
             car = paste(manufacturer, model, sep = " "),
             `cyl / trans` = paste(cyl, " cylinders", " / ", trans, " transmission", sep = ""))
dados
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
```

```
## # ... with 224 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

2.5 transmute()

```
transmute(dados,
          `avg miles per gallon` = (cty + hwy) / 2)
```

```
## # A tibble: 234 x 1
##    `avg miles per gallon`
##                     <dbl>
##  1                   23.5
##  2                   25
##  3                   25.5
##  4                   25.5
##  5                   21
##  6                   22
##  7                   22.5
##  8                   22
##  9                   20.5
## 10                   24
## # ... with 224 more rows
```

```
transmute(dados,
          car = paste(manufacturer, model, sep = " "),
          `cyl / trans` = paste(cyl, " cylinders", " / ", trans, " transmission", sep =
```

```
## # A tibble: 234 x 2
##    car             `cyl / trans`
##    <chr>           <chr>
##  1 audi a4         4 cylinders / auto(l5) transmission
##  2 audi a4         4 cylinders / manual(m5) transmission
##  3 audi a4         4 cylinders / manual(m6) transmission
##  4 audi a4         4 cylinders / auto(av) transmission
##  5 audi a4         6 cylinders / auto(l5) transmission
##  6 audi a4         6 cylinders / manual(m5) transmission
##  7 audi a4         6 cylinders / auto(av) transmission
##  8 audi a4 quattro 4 cylinders / manual(m5) transmission
##  9 audi a4 quattro 4 cylinders / auto(l5) transmission
## 10 audi a4 quattro 4 cylinders / manual(m6) transmission
## # ... with 224 more rows
```

2.6 filter() para linhas

```
filter(dados, manufacturer == "audi")
```

```
## # A tibble: 18 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## 11 audi      a4 q~   2    2008     4 auto~ 4        19    27 p     comp~    23
## 12 audi      a4 q~   2.8  1999     6 auto~ 4        15    25 p     comp~    20
## 13 audi      a4 q~   2.8  1999     6 manu~ 4        17    25 p     comp~    21
## 14 audi      a4 q~   3.1  2008     6 auto~ 4        17    25 p     comp~    21
## 15 audi      a4 q~   3.1  2008     6 manu~ 4        15    25 p     comp~    20
## 16 audi      a6 q~   2.8  1999     6 auto~ 4        15    24 p     mids~    19.5
## 17 audi      a6 q~   3.1  2008     6 auto~ 4        17    25 p     mids~    21
## 18 audi      a6 q~   4.2  2008     8 auto~ 4        16    23 p     mids~    19.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, manufacturer == "audi" & year == 1999)
```

```
## # A tibble: 9 x 14
##   manufact~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##   <chr>      <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
## 1 audi       a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
## 2 audi       a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
## 3 audi       a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
## 4 audi       a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
## 5 audi       a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
## 6 audi       a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 7 audi       a4 q~   2.8  1999     6 auto~ 4        15    25 p     comp~    20
## 8 audi       a4 q~   2.8  1999     6 manu~ 4        17    25 p     comp~    21
## 9 audi       a6 q~   2.8  1999     6 auto~ 4        15    24 p     mids~    19.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```r
filter(dados, manufacturer == "audi", year == 1999)
```

```
## # A tibble: 9 x 14
##   manufact~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##   <chr>      <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
## 1 audi       a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
## 2 audi       a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
## 3 audi       a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
## 4 audi       a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
## 5 audi       a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
## 6 audi       a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 7 audi       a4 q~   2.8  1999     6 auto~ 4        15    25 p     comp~    20
## 8 audi       a4 q~   2.8  1999     6 manu~ 4        17    25 p     comp~    21
## 9 audi       a6 q~   2.8  1999     6 auto~ 4        15    24 p     mids~    19.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```r
filter(dados, manufacturer == "audi" | manufacturer == "dodge")
```

```
## # A tibble: 55 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 45 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```r
filter(dados, manufacturer %in% c("audi", "dodge"))
```

```
## # A tibble: 55 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
```

```
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 45 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, hwy >= 30)
```

```
## # A tibble: 26 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  2 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  3 chevrolet mali~   2.4  2008     4 auto~ f        22    30 r     mids~    26
##  4 honda     civic   1.6  1999     4 manu~ f        28    33 r     subc~    30.5
##  5 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  6 honda     civic   1.6  1999     4 manu~ f        25    32 r     subc~    28.5
##  7 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  8 honda     civic   1.8  2008     4 manu~ f        26    34 r     subc~    30
##  9 honda     civic   1.8  2008     4 auto~ f        25    36 r     subc~    30.5
## 10 honda     civic   1.8  2008     4 auto~ f        24    36 c     subc~    30
## # ... with 16 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, year != 1999)
```

```
## # A tibble: 117 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  2 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  3 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  4 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
##  5 audi      a4 q~   2    2008     4 auto~ 4        19    27 p     comp~    23
##  6 audi      a4 q~   3.1  2008     6 auto~ 4        17    25 p     comp~    21
##  7 audi      a4 q~   3.1  2008     6 manu~ 4        15    25 p     comp~    20
##  8 audi      a6 q~   3.1  2008     6 auto~ 4        17    25 p     mids~    21
##  9 audi      a6 q~   4.2  2008     8 auto~ 4        16    23 p     mids~    19.5
## 10 chevrolet c150~   5.3  2008     8 auto~ r        14    20 r     suv      17
## # ... with 107 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, between(cty,15, 22))
```

```
## # A tibble: 143 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
```

```
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 133 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

2.7 slice() para linhas

```
slice(dados, 1:5)
```

```
## # A tibble: 5 x 14
##   manufact~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##   <chr>      <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
## 1 audi       a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
## 2 audi       a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
## 3 audi       a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
## 4 audi       a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
## 5 audi       a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
dados[1:5,]
```

```
## # A tibble: 5 x 14
##   manufact~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##   <chr>      <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
## 1 audi       a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
## 2 audi       a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
## 3 audi       a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
## 4 audi       a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
## 5 audi       a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
slice(dados, 20:30)
```

```
## # A tibble: 11 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 chevrolet c150~   5.3  2008     8 auto~ r        11    15 e     suv      13
```

```
##  2 chevrolet c150~   5.3  2008     8 auto~ r        14    20 r     suv      17
##  3 chevrolet c150~   5.7  1999     8 auto~ r        13    17 r     suv      15
##  4 chevrolet c150~   6    2008     8 auto~ r        12    17 r     suv      14.5
##  5 chevrolet corv~   5.7  1999     8 manu~ r        16    26 p     2sea~    21
##  6 chevrolet corv~   5.7  1999     8 auto~ r        15    23 p     2sea~    19
##  7 chevrolet corv~   6.2  2008     8 manu~ r        16    26 p     2sea~    21
##  8 chevrolet corv~   6.2  2008     8 auto~ r        15    25 p     2sea~    20
##  9 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
## 10 chevrolet k150~   5.3  2008     8 auto~ 4        14    19 r     suv      16.5
## 11 chevrolet k150~   5.3  2008     8 auto~ 4        11    14 e     suv      12.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
dados[20:30,]
```

```
## # A tibble: 11 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 chevrolet c150~   5.3  2008     8 auto~ r        11    15 e     suv      13
##  2 chevrolet c150~   5.3  2008     8 auto~ r        14    20 r     suv      17
##  3 chevrolet c150~   5.7  1999     8 auto~ r        13    17 r     suv      15
##  4 chevrolet c150~   6    2008     8 auto~ r        12    17 r     suv      14.5
##  5 chevrolet corv~   5.7  1999     8 manu~ r        16    26 p     2sea~    21
##  6 chevrolet corv~   5.7  1999     8 auto~ r        15    23 p     2sea~    19
##  7 chevrolet corv~   6.2  2008     8 manu~ r        16    26 p     2sea~    21
##  8 chevrolet corv~   6.2  2008     8 auto~ r        15    25 p     2sea~    20
##  9 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
## 10 chevrolet k150~   5.3  2008     8 auto~ 4        14    19 r     suv      16.5
## 11 chevrolet k150~   5.3  2008     8 auto~ 4        11    14 e     suv      12.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

2.8 arrange() para linhas

```
arrange(dados, displ)
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 honda     civic   1.6  1999     4 manu~ f        28    33 r     subc~    30.5
##  2 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  3 honda     civic   1.6  1999     4 manu~ f        25    32 r     subc~    28.5
##  4 honda     civic   1.6  1999     4 manu~ f        23    29 p     subc~    26
##  5 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  6 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  7 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
```

```
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 honda     civic   1.8  2008     4 manu~ f        26    34 r     subc~    30
## # ... with 224 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
arrange(dados, displ) %>%
  print(n=20)
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 honda     civic   1.6  1999     4 manu~ f        28    33 r     subc~    30.5
##  2 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  3 honda     civic   1.6  1999     4 manu~ f        25    32 r     subc~    28.5
##  4 honda     civic   1.6  1999     4 manu~ f        23    29 p     subc~    26
##  5 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  6 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  7 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 honda     civic   1.8  2008     4 manu~ f        26    34 r     subc~    30
## 11 honda     civic   1.8  2008     4 auto~ f        25    36 r     subc~    30.5
## 12 honda     civic   1.8  2008     4 auto~ f        24    36 c     subc~    30
## 13 toyota    coro~   1.8  1999     4 auto~ f        24    30 r     comp~    27
## 14 toyota    coro~   1.8  1999     4 auto~ f        24    33 r     comp~    28.5
## 15 toyota    coro~   1.8  1999     4 manu~ f        26    35 r     comp~    30.5
## 16 toyota    coro~   1.8  2008     4 manu~ f        28    37 r     comp~    32.5
## 17 toyota    coro~   1.8  2008     4 auto~ f        26    35 r     comp~    30.5
## 18 volkswag~ pass~   1.8  1999     4 manu~ f        21    29 p     mids~    25
## 19 volkswag~ pass~   1.8  1999     4 auto~ f        18    29 p     mids~    23.5
## 20 volkswag~ jetta   1.9  1999     4 manu~ f        33    44 d     comp~    38.5
## # ... with 214 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
arrange(dados, desc(displ))
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
##  2 chevrolet k150~   6.5  1999     8 auto~ 4        14    17 d     suv      15.5
##  3 chevrolet corv~   6.2  2008     8 manu~ r        16    26 p     2sea~    21
##  4 chevrolet corv~   6.2  2008     8 auto~ r        15    25 p     2sea~    20
##  5 jeep      gran~   6.1  2008     8 auto~ 4        11    14 p     suv      12.5
##  6 chevrolet c150~   6    2008     8 auto~ r        12    17 r     suv      14.5
##  7 dodge     dura~   5.9  1999     8 auto~ 4        11    15 r     suv      13
```

```
## 8 dodge     ram ~   5.9  1999     8 auto~ 4        11    15 r     pick~    13
## 9 chevrolet c150~   5.7  1999     8 auto~ r        13    17 r     suv      15
## 10 chevrolet corv~   5.7  1999     8 manu~ r        16    26 p     2sea~    21
## # ... with 224 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
arrange(dados, desc(displ)) %>%
  print(n=20)
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
##  2 chevrolet k150~   6.5  1999     8 auto~ 4        14    17 d     suv      15.5
##  3 chevrolet corv~   6.2  2008     8 manu~ r        16    26 p     2sea~    21
##  4 chevrolet corv~   6.2  2008     8 auto~ r        15    25 p     2sea~    20
##  5 jeep      gran~   6.1  2008     8 auto~ 4        11    14 p     suv      12.5
##  6 chevrolet c150~   6    2008     8 auto~ r        12    17 r     suv      14.5
##  7 dodge     dura~   5.9  1999     8 auto~ 4        11    15 r     suv      13
##  8 dodge     ram ~   5.9  1999     8 auto~ 4        11    15 r     pick~    13
##  9 chevrolet c150~   5.7  1999     8 auto~ r        13    17 r     suv      15
## 10 chevrolet corv~   5.7  1999     8 manu~ r        16    26 p     2sea~    21
## 11 chevrolet corv~   5.7  1999     8 auto~ r        15    23 p     2sea~    19
## 12 chevrolet k150~   5.7  1999     8 auto~ 4        11    15 r     suv      13
## 13 dodge     dura~   5.7  2008     8 auto~ 4        13    18 r     suv      15.5
## 14 dodge     ram ~   5.7  2008     8 auto~ 4        13    17 r     pick~    15
## 15 jeep      gran~   5.7  2008     8 auto~ 4        13    18 r     suv      15.5
## 16 toyota    land~   5.7  2008     8 auto~ 4        13    18 r     suv      15.5
## 17 nissan    path~   5.6  2008     8 auto~ 4        12    18 p     suv      15
## 18 ford      expe~   5.4  1999     8 auto~ r        11    17 r     suv      14
## 19 ford      expe~   5.4  2008     8 auto~ r        12    18 r     suv      15
## 20 ford      f150~   5.4  1999     8 auto~ 4        11    15 r     pick~    13
## # ... with 214 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
select(dados, displ, cty) %>%
  arrange(displ, cty) %>%
  print(n = 20)
```

```
## # A tibble: 234 x 2
##    displ   cty
##    <dbl> <int>
##  1   1.6    23
##  2   1.6    24
##  3   1.6    24
##  4   1.6    25
##  5   1.6    28
```

```
##  6   1.8    16
##  7   1.8    18
##  8   1.8    18
##  9   1.8    18
## 10   1.8    21
## 11   1.8    21
## 12   1.8    24
## 13   1.8    24
## 14   1.8    24
## 15   1.8    25
## 16   1.8    26
## 17   1.8    26
## 18   1.8    26
## 19   1.8    28
## 20   1.9    29
## # ... with 214 more rows
```

2.9 distinct() para linhas

```
dados_exemplo <- data.frame(id = 1:3,
                            name = c("John", "Max", "Julia"))
dados_exemplo
```

```
##   id  name
## 1  1  John
## 2  2   Max
## 3  3 Julia
```

```
# bind_rows == rbind()
dados_exemplo<- bind_rows(dados_exemplo, slice(dados_exemplo, 2))
dados_exemplo
```

```
##   id  name
## 1  1  John
## 2  2   Max
## 3  3 Julia
## 4  2   Max
```

```
distinct(dados_exemplo)
```

```
##   id  name
## 1  1  John
## 2  2   Max
## 3  3 Julia
```

```
dados_exemplo2 <- data.frame(id = c(1,1,2),
                            name = c("John", "Max", "Julia"))
```

```
dados_exemplo2
```

```
##   id  name
## 1  1  John
## 2  1   Max
## 3  2 Julia
```

```
distinct(dados_exemplo2)
```

```
##   id  name
## 1  1  John
## 2  1   Max
## 3  2 Julia
```

```
dados_duplicados <- select(dados, manufacturer, model)
dados_duplicados
```

```
## # A tibble: 234 x 2
##    manufacturer model
##    <chr>        <chr>
##  1 audi         a4
##  2 audi         a4
##  3 audi         a4
##  4 audi         a4
##  5 audi         a4
##  6 audi         a4
##  7 audi         a4
##  8 audi         a4 quattro
##  9 audi         a4 quattro
## 10 audi         a4 quattro
## # ... with 224 more rows
```

```
dados_nao_duplicados <- distinct(dados_duplicados)
dados_nao_duplicados
```

```
## # A tibble: 38 x 2
##    manufacturer model
##    <chr>        <chr>
##  1 audi         a4
##  2 audi         a4 quattro
##  3 audi         a6 quattro
##  4 chevrolet    c1500 suburban 2wd
##  5 chevrolet    corvette
##  6 chevrolet    k1500 tahoe 4wd
##  7 chevrolet    malibu
##  8 dodge        caravan 2wd
##  9 dodge        dakota pickup 4wd
## 10 dodge        durango 4wd
```

```
## # ... with 28 more rows
```

2.10 summarise()

```
summarise(dados, `mean hwy` = mean(hwy))
```

```
## # A tibble: 1 x 1
##   `mean hwy`
##        <dbl>
## 1       23.4
```

```
summarise(dados,
          rows = n(),
          `nr models` = n_distinct(model))
```

```
## # A tibble: 1 x 2
##    rows `nr models`
##   <int>       <int>
## 1   234          38
```

```
summarise(dados,
          `min hwy` = min(hwy),
          `min cty` = min(cty),
          `max hwy` = max(hwy),
          `max cty` = max(cty))
```

```
## # A tibble: 1 x 4
##   `min hwy` `min cty` `max hwy` `max cty`
##       <int>     <int>     <int>     <int>
## 1        12         9        44        35
```

2.11 group_by()

```
group_by(dados, manufacturer)
```

```
## # A tibble: 234 x 14
## # Groups:   manufacturer [15]
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
```

```
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 224 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
dados %>%
  group_by(manufacturer) %>%
  summarise(cars = n())
```

```
## # A tibble: 15 x 2
##    manufacturer  cars
##    <chr>        <int>
##  1 audi            18
##  2 chevrolet       19
##  3 dodge           37
##  4 ford            25
##  5 honda            9
##  6 hyundai         14
##  7 jeep             8
##  8 land rover       4
##  9 lincoln          3
## 10 mercury          4
## 11 nissan          13
## 12 pontiac          5
## 13 subaru          14
## 14 toyota          34
## 15 volkswagen      27
```

```
dados %>%
  group_by(model) %>%
  summarise(`mean hwy` = mean(hwy),
            `min hwy` = min(hwy),
            `max hwy` = max(hwy))
```

```
## # A tibble: 38 x 4
##    model               `mean hwy` `min hwy` `max hwy`
##    <chr>                    <dbl>     <int>     <int>
##  1 4runner 4wd               18.8        17        20
##  2 a4                        28.3        26        31
##  3 a4 quattro                25.8        25        28
##  4 a6 quattro                24          23        25
##  5 altima                    28.7        26        32
##  6 c1500 suburban 2wd        17.8        15        20
##  7 camry                     28.3        26        31
##  8 camry solara              28.1        26        31
```

```
##  9 caravan 2wd              22.4        17        24
## 10 civic                    32.6        29        36
## # ... with 28 more rows
```

2.12 count()

```
count(dados)
```

```
## # A tibble: 1 x 1
##       n
##   <int>
## 1   234
```

```
dados %>%
  group_by(model) %>%
  count()
```

```
## # A tibble: 38 x 2
## # Groups:   model [38]
##    model                  n
##    <chr>              <int>
##  1 4runner 4wd            6
##  2 a4                     7
##  3 a4 quattro             8
##  4 a6 quattro             3
##  5 altima                 6
##  6 c1500 suburban 2wd     5
##  7 camry                  7
##  8 camry solara           7
##  9 caravan 2wd           11
## 10 civic                  9
## # ... with 28 more rows
```

2.13 sample_n()

```
set.seed(567)
sample_n(dados, size = 10, replace = F)
```

```
## # A tibble: 10 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
##  1 mercury   moun~   5    1999     8 auto~ 4        13    17 r     suv      15
##  2 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
##  3 dodge     ram ~   4.7  2008     8 manu~ 4        12    16 r     pick~    14
##  4 toyota    land~   4.7  1999     8 auto~ 4        11    15 r     suv      13
```

```
## 5 volkswag~ jetta   2    1999     4 auto~ f        19    26 r     comp~    22.5
## 6 dodge     cara~   3.8  1999     6 auto~ f        15    21 r     mini~    18
## 7 honda     civic   1.8  2008     4 auto~ f        25    36 r     subc~    30.5
## 8 ford      must~   4.6  1999     8 auto~ r        15    21 r     subc~    18
## 9 chevrolet c150~   5.3  2008     8 auto~ r        14    20 r     suv      17
## 10 ford     expe~   5.4  1999     8 auto~ r        11    17 r     suv      14
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
sample_n(dados, size = 10, replace = T)
```

```
## # A tibble: 10 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
## 1 chevrolet c150~   5.3  2008     8 auto~ r        11    15 e     suv      13
## 2 volkswag~ gti     2    2008     4 auto~ f        22    29 p     comp~    25.5
## 3 dodge     dako~   4.7  2008     8 auto~ 4        14    19 r     pick~    16.5
## 4 ford      expl~   4.6  2008     8 auto~ 4        13    19 r     suv      16
## 5 dodge     cara~   3.8  2008     6 auto~ f        16    23 r     mini~    19.5
## 6 chevrolet k150~   5.3  2008     8 auto~ 4        14    19 r     suv      16.5
## 7 dodge     dura~   5.2  1999     8 auto~ 4        11    16 r     suv      13.5
## 8 toyota    camry   2.4  2008     4 manu~ f        21    31 r     mids~    26
## 9 toyota    camry   3    1999     6 manu~ f        18    26 r     mids~    22
## 10 subaru   impr~   2.2  1999     4 auto~ 4        21    26 r     subc~    23.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

2.14 sample_frac()

```
sample_frac(dados, size = 0.1, replace = F)
```

```
## # A tibble: 23 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <chr>     <chr> <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>   <dbl>
## 1 toyota    coro~   1.8  2008     4 manu~ f        28    37 r     comp~    32.5
## 2 lincoln   navi~   5.4  1999     8 auto~ r        11    17 r     suv      14
## 3 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
## 4 audi      a6 q~   2.8  1999     6 auto~ 4        15    24 p     mids~    19.5
## 5 nissan    path~   4    2008     6 auto~ 4        14    20 p     suv      17
## 6 toyota    camry   3.5  2008     6 auto~ f        19    28 r     mids~    23.5
## 7 subaru    impr~   2.5  2008     4 auto~ 4        20    25 p     comp~    22.5
## 8 toyota    toyo~   3.4  1999     6 auto~ 4        15    19 r     pick~    17
## 9 audi      a4 q~   3.1  2008     6 manu~ 4        15    25 p     comp~    20
## 10 toyota   coro~   1.8  1999     4 manu~ f        26    35 r     comp~    30.5
## # ... with 13 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
```

```
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

Chapter 3

ggplot2 (60 minutos)

Alguns links

The R Graph Gallery

120 registered extensions available to explore

link 1: patchwork

link 2: patchwork

```
ls("package:ggplot2")
```

```
##   [1] "%+%"                       "%+replace%"
##   [3] "aes"                       "aes_"
##   [5] "aes_all"                   "aes_auto"
##   [7] "aes_q"                     "aes_string"
##   [9] "after_scale"               "after_stat"
##  [11] "alpha"                     "annotate"
##  [13] "annotation_custom"         "annotation_logticks"
##  [15] "annotation_map"            "annotation_raster"
##  [17] "arrow"                     "as_label"
##  [19] "as_labeller"               "autolayer"
##  [21] "autoplot"                  "AxisSecondary"
##  [23] "benchplot"                 "binned_scale"
##  [25] "borders"                   "calc_element"
##  [27] "combine_vars"              "continuous_scale"
##  [29] "Coord"                     "coord_cartesian"
##  [31] "coord_equal"               "coord_fixed"
##  [33] "coord_flip"                "coord_map"
##  [35] "coord_munch"               "coord_polar"
##  [37] "coord_quickmap"            "coord_sf"
##  [39] "coord_trans"               "CoordCartesian"
```

```
##  [41] "CoordFixed"                "CoordFlip"
##  [43] "CoordMap"                  "CoordPolar"
##  [45] "CoordQuickmap"             "CoordSf"
##  [47] "CoordTrans"                "cut_interval"
##  [49] "cut_number"                "cut_width"
##  [51] "derive"                    "diamonds"
##  [53] "discrete_scale"            "draw_key_abline"
##  [55] "draw_key_blank"            "draw_key_boxplot"
##  [57] "draw_key_crossbar"         "draw_key_dotplot"
##  [59] "draw_key_label"            "draw_key_path"
##  [61] "draw_key_point"            "draw_key_pointrange"
##  [63] "draw_key_polygon"          "draw_key_rect"
##  [65] "draw_key_smooth"           "draw_key_text"
##  [67] "draw_key_timeseries"       "draw_key_vline"
##  [69] "draw_key_vpath"            "dup_axis"
##  [71] "economics"                 "economics_long"
##  [73] "el_def"                    "element_blank"
##  [75] "element_grob"              "element_line"
##  [77] "element_rect"              "element_render"
##  [79] "element_text"              "enexpr"
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##  [83] "enquos"                    "ensym"
##  [85] "ensyms"                    "expand_limits"
##  [87] "expand_scale"              "expansion"
##  [89] "expr"                      "Facet"
##  [91] "facet_grid"                "facet_null"
##  [93] "facet_wrap"                "FacetGrid"
##  [95] "FacetNull"                 "FacetWrap"
##  [97] "faithfuld"                 "find_panel"
##  [99] "flip_data"                 "flipped_names"
## [101] "fortify"                   "Geom"
## [103] "geom_abline"               "geom_area"
## [105] "geom_bar"                  "geom_bin_2d"
## [107] "geom_bin2d"                "geom_blank"
## [109] "geom_boxplot"              "geom_col"
## [111] "geom_contour"              "geom_contour_filled"
## [113] "geom_count"                "geom_crossbar"
## [115] "geom_curve"                "geom_density"
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## [119] "geom_density2d"            "geom_density2d_filled"
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## [123] "geom_errorbarh"            "geom_freqpoly"
## [125] "geom_function"             "geom_hex"
## [127] "geom_histogram"            "geom_hline"
## [129] "geom_jitter"               "geom_label"
## [131] "geom_line"                 "geom_linerange"
```

```
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## [135] "geom_point"                "geom_pointrange"
## [137] "geom_polygon"              "geom_qq"
## [139] "geom_qq_line"              "geom_quantile"
## [141] "geom_raster"               "geom_rect"
## [143] "geom_ribbon"               "geom_rug"
## [145] "geom_segment"              "geom_sf"
## [147] "geom_sf_label"             "geom_sf_text"
## [149] "geom_smooth"               "geom_spoke"
## [151] "geom_step"                 "geom_text"
## [153] "geom_tile"                 "geom_violin"
## [155] "geom_vline"                "GeomAbline"
## [157] "GeomAnnotationMap"         "GeomArea"
## [159] "GeomBar"                   "GeomBlank"
## [161] "GeomBoxplot"               "GeomCol"
## [163] "GeomContour"               "GeomContourFilled"
## [165] "GeomCrossbar"              "GeomCurve"
## [167] "GeomCustomAnn"             "GeomDensity"
## [169] "GeomDensity2d"             "GeomDensity2dFilled"
## [171] "GeomDotplot"               "GeomErrorbar"
## [173] "GeomErrorbarh"             "GeomFunction"
## [175] "GeomHex"                   "GeomHline"
## [177] "GeomLabel"                 "GeomLine"
## [179] "GeomLinerange"             "GeomLogticks"
## [181] "GeomMap"                   "GeomPath"
## [183] "GeomPoint"                 "GeomPointrange"
## [185] "GeomPolygon"               "GeomQuantile"
## [187] "GeomRaster"                "GeomRasterAnn"
## [189] "GeomRect"                  "GeomRibbon"
## [191] "GeomRug"                   "GeomSegment"
## [193] "GeomSf"                    "GeomSmooth"
## [195] "GeomSpoke"                 "GeomStep"
## [197] "GeomText"                  "GeomTile"
## [199] "GeomViolin"                "GeomVline"
## [201] "get_alt_text"              "get_element_tree"
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## [205] "ggplot_add"                "ggplot_build"
## [207] "ggplot_gtable"             "ggplotGrob"
## [209] "ggproto"                   "ggproto_parent"
## [211] "ggsave"                    "ggtitle"
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## [215] "guide_colorbar"            "guide_colorsteps"
## [217] "guide_colourbar"           "guide_coloursteps"
## [219] "guide_gengrob"             "guide_geom"
## [221] "guide_legend"              "guide_merge"
## [223] "guide_none"                "guide_train"
```

```
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## [227] "has_flipped_aes"          "is.Coord"
## [229] "is.facet"                 "is.ggplot"
## [231] "is.ggproto"               "is.theme"
## [233] "label_both"               "label_bquote"
## [235] "label_context"            "label_parsed"
## [237] "label_value"              "label_wrap_gen"
## [239] "labeller"                 "labs"
## [241] "last_plot"                "layer"
## [243] "layer_data"               "layer_grob"
## [245] "layer_scales"             "layer_sf"
## [247] "Layout"                   "lims"
## [249] "luv_colours"              "map_data"
## [251] "margin"                   "max_height"
## [253] "max_width"                "mean_cl_boot"
## [255] "mean_cl_normal"           "mean_sdl"
## [257] "mean_se"                  "median_hilow"
## [259] "merge_element"            "midwest"
## [261] "mpg"                      "msleep"
## [263] "panel_cols"               "panel_rows"
## [265] "Position"                 "position_dodge"
## [267] "position_dodge2"          "position_fill"
## [269] "position_identity"        "position_jitter"
## [271] "position_jitterdodge"     "position_nudge"
## [273] "position_stack"           "PositionDodge"
## [275] "PositionDodge2"           "PositionFill"
## [277] "PositionIdentity"         "PositionJitter"
## [279] "PositionJitterdodge"      "PositionNudge"
## [281] "PositionStack"            "presidential"
## [283] "qplot"                    "quickplot"
## [285] "quo"                      "quo_name"
## [287] "quos"                     "register_theme_elements"
## [289] "rel"                      "remove_missing"
## [291] "render_axes"              "render_strips"
## [293] "reset_theme_settings"     "resolution"
## [295] "Scale"                    "scale_alpha"
## [297] "scale_alpha_binned"       "scale_alpha_continuous"
## [299] "scale_alpha_date"         "scale_alpha_datetime"
## [301] "scale_alpha_discrete"     "scale_alpha_identity"
## [303] "scale_alpha_manual"       "scale_alpha_ordinal"
## [305] "scale_color_binned"       "scale_color_brewer"
## [307] "scale_color_continuous"   "scale_color_date"
## [309] "scale_color_datetime"     "scale_color_discrete"
## [311] "scale_color_distiller"    "scale_color_fermenter"
## [313] "scale_color_gradient"     "scale_color_gradient2"
## [315] "scale_color_gradientn"    "scale_color_grey"
```

```
## [317] "scale_color_hue"           "scale_color_identity"
## [319] "scale_color_manual"        "scale_color_ordinal"
## [321] "scale_color_steps"         "scale_color_steps2"
## [323] "scale_color_stepsn"        "scale_color_viridis_b"
## [325] "scale_color_viridis_c"     "scale_color_viridis_d"
## [327] "scale_colour_binned"       "scale_colour_brewer"
## [329] "scale_colour_continuous"   "scale_colour_date"
## [331] "scale_colour_datetime"     "scale_colour_discrete"
## [333] "scale_colour_distiller"    "scale_colour_fermenter"
## [335] "scale_colour_gradient"     "scale_colour_gradient2"
## [337] "scale_colour_gradientn"    "scale_colour_grey"
## [339] "scale_colour_hue"          "scale_colour_identity"
## [341] "scale_colour_manual"       "scale_colour_ordinal"
## [343] "scale_colour_steps"        "scale_colour_steps2"
## [345] "scale_colour_stepsn"       "scale_colour_viridis_b"
## [347] "scale_colour_viridis_c"    "scale_colour_viridis_d"
## [349] "scale_continuous_identity" "scale_discrete_identity"
## [351] "scale_discrete_manual"     "scale_fill_binned"
## [353] "scale_fill_brewer"         "scale_fill_continuous"
## [355] "scale_fill_date"           "scale_fill_datetime"
## [357] "scale_fill_discrete"       "scale_fill_distiller"
## [359] "scale_fill_fermenter"      "scale_fill_gradient"
## [361] "scale_fill_gradient2"      "scale_fill_gradientn"
## [363] "scale_fill_grey"           "scale_fill_hue"
## [365] "scale_fill_identity"       "scale_fill_manual"
## [367] "scale_fill_ordinal"        "scale_fill_steps"
## [369] "scale_fill_steps2"         "scale_fill_stepsn"
## [371] "scale_fill_viridis_b"      "scale_fill_viridis_c"
## [373] "scale_fill_viridis_d"      "scale_linetype"
## [375] "scale_linetype_binned"     "scale_linetype_continuous"
## [377] "scale_linetype_discrete"   "scale_linetype_identity"
## [379] "scale_linetype_manual"     "scale_radius"
## [381] "scale_shape"               "scale_shape_binned"
## [383] "scale_shape_continuous"    "scale_shape_discrete"
## [385] "scale_shape_identity"      "scale_shape_manual"
## [387] "scale_shape_ordinal"       "scale_size"
## [389] "scale_size_area"           "scale_size_binned"
## [391] "scale_size_binned_area"    "scale_size_continuous"
## [393] "scale_size_date"           "scale_size_datetime"
## [395] "scale_size_discrete"       "scale_size_identity"
## [397] "scale_size_manual"         "scale_size_ordinal"
## [399] "scale_type"                "scale_x_binned"
## [401] "scale_x_continuous"        "scale_x_date"
## [403] "scale_x_datetime"          "scale_x_discrete"
## [405] "scale_x_log10"             "scale_x_reverse"
## [407] "scale_x_sqrt"              "scale_x_time"
```

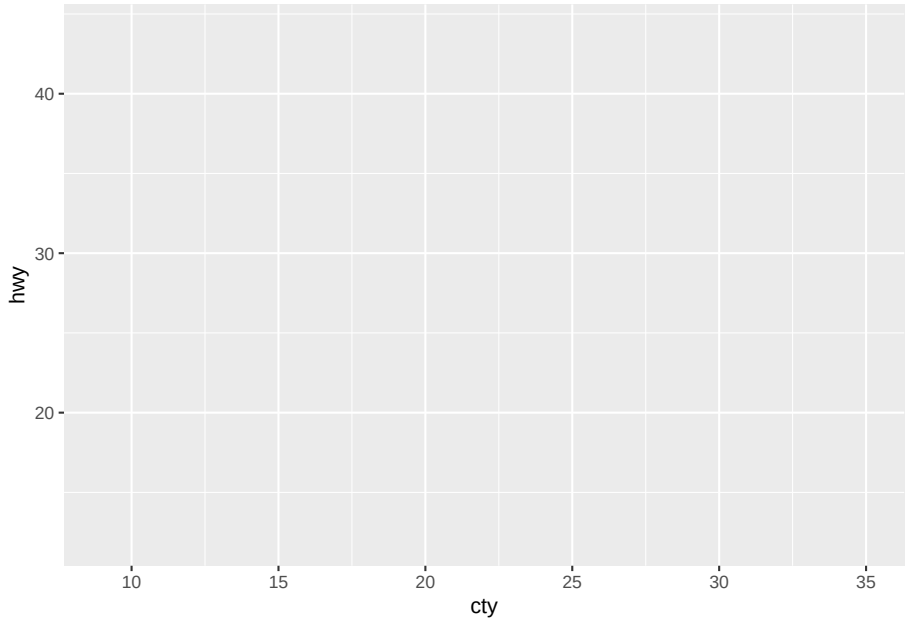
```
## [409] "scale_y_binned"            "scale_y_continuous"
## [411] "scale_y_date"              "scale_y_datetime"
## [413] "scale_y_discrete"          "scale_y_log10"
## [415] "scale_y_reverse"           "scale_y_sqrt"
## [417] "scale_y_time"              "ScaleBinned"
## [419] "ScaleBinnedPosition"       "ScaleContinuous"
## [421] "ScaleContinuousDate"       "ScaleContinuousDatetime"
## [423] "ScaleContinuousIdentity"   "ScaleContinuousPosition"
## [425] "ScaleDiscrete"             "ScaleDiscreteIdentity"
## [427] "ScaleDiscretePosition"     "seals"
## [429] "sec_axis"                  "set_last_plot"
## [431] "sf_transform_xy"           "should_stop"
## [433] "stage"                     "standardise_aes_names"
## [435] "stat"                      "Stat"
## [437] "stat_bin"                  "stat_bin_2d"
## [439] "stat_bin_hex"              "stat_bin2d"
## [441] "stat_binhex"               "stat_boxplot"
## [443] "stat_contour"              "stat_contour_filled"
## [445] "stat_count"                "stat_density"
## [447] "stat_density_2d"           "stat_density_2d_filled"
## [449] "stat_density2d"            "stat_density2d_filled"
## [451] "stat_ecdf"                 "stat_ellipse"
## [453] "stat_function"             "stat_identity"
## [455] "stat_qq"                   "stat_qq_line"
## [457] "stat_quantile"             "stat_sf"
## [459] "stat_sf_coordinates"       "stat_smooth"
## [461] "stat_spoke"                "stat_sum"
## [463] "stat_summary"              "stat_summary_2d"
## [465] "stat_summary_bin"          "stat_summary_hex"
## [467] "stat_summary2d"            "stat_unique"
## [469] "stat_ydensity"             "StatBin"
## [471] "StatBin2d"                 "StatBindot"
## [473] "StatBinhex"                "StatBoxplot"
## [475] "StatContour"               "StatContourFilled"
## [477] "StatCount"                 "StatDensity"
## [479] "StatDensity2d"             "StatDensity2dFilled"
## [481] "StatEcdf"                  "StatEllipse"
## [483] "StatFunction"              "StatIdentity"
## [485] "StatQq"                    "StatQqLine"
## [487] "StatQuantile"              "StatSf"
## [489] "StatSfCoordinates"         "StatSmooth"
## [491] "StatSum"                   "StatSummary"
## [493] "StatSummary2d"             "StatSummaryBin"
## [495] "StatSummaryHex"            "StatUnique"
## [497] "StatYdensity"              "summarise_coord"
## [499] "summarise_layers"          "summarise_layout"
```

```
## [501] "sym"                      "syms"
## [503] "theme"                    "theme_bw"
## [505] "theme_classic"            "theme_dark"
## [507] "theme_get"                "theme_gray"
## [509] "theme_grey"               "theme_light"
## [511] "theme_linedraw"           "theme_minimal"
## [513] "theme_replace"            "theme_set"
## [515] "theme_test"               "theme_update"
## [517] "theme_void"               "transform_position"
## [519] "txhousing"                "unit"
## [521] "update_geom_defaults"     "update_labels"
## [523] "update_stat_defaults"     "vars"
## [525] "waiver"                   "wrap_dims"
## [527] "xlab"                     "xlim"
## [529] "ylab"                     "ylim"
## [531] "zeroGrob"
```

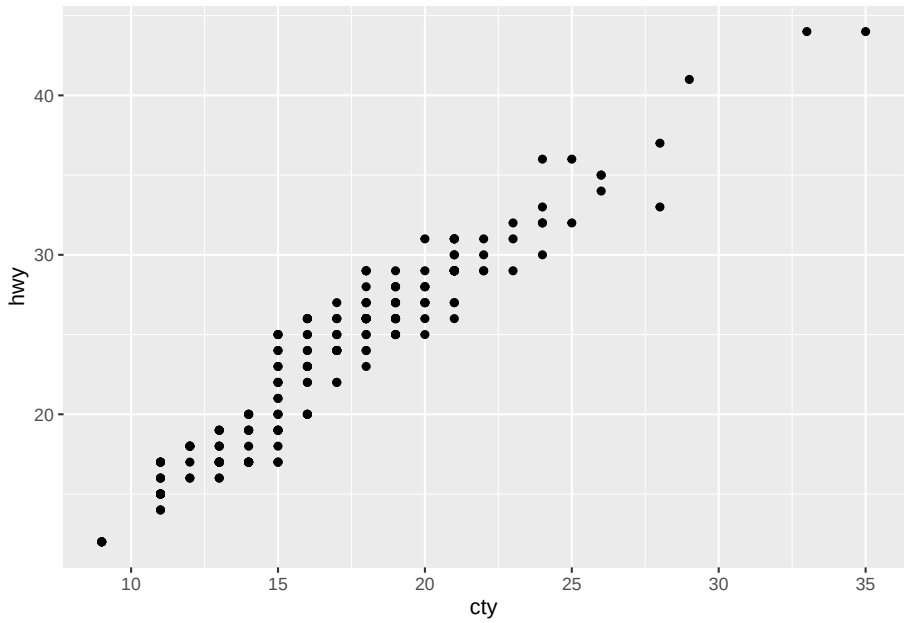
3.1 Primeiros passos

```
ggplot(dados)
```

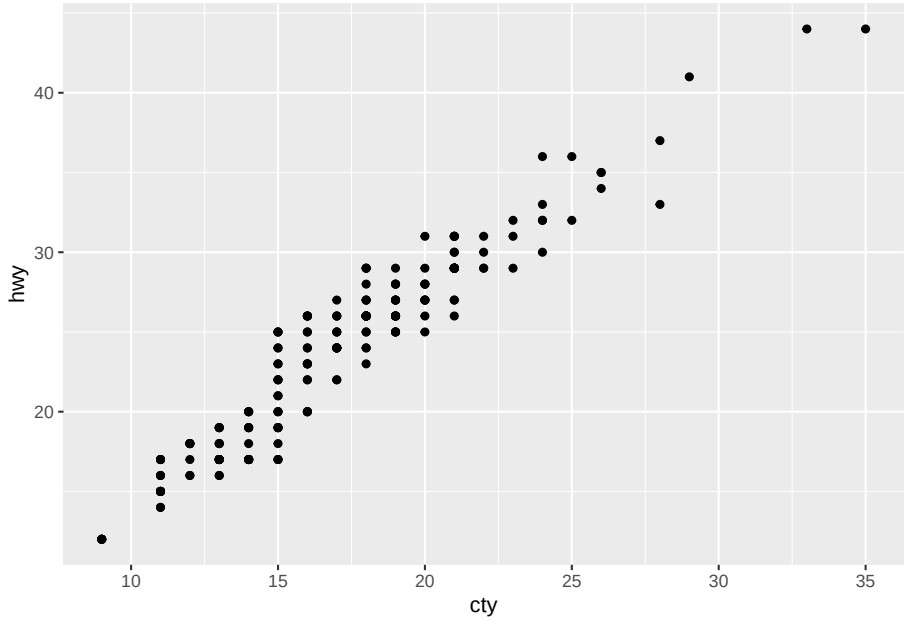
```
ggplot(dados, aes(x = cty, y = hwy))
```



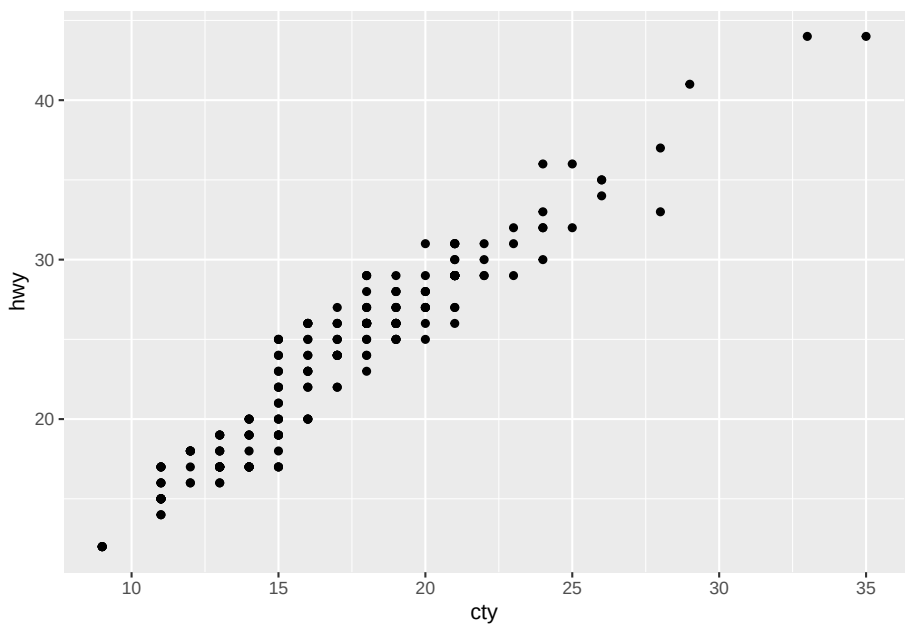

```
# Alternativas
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point()
```



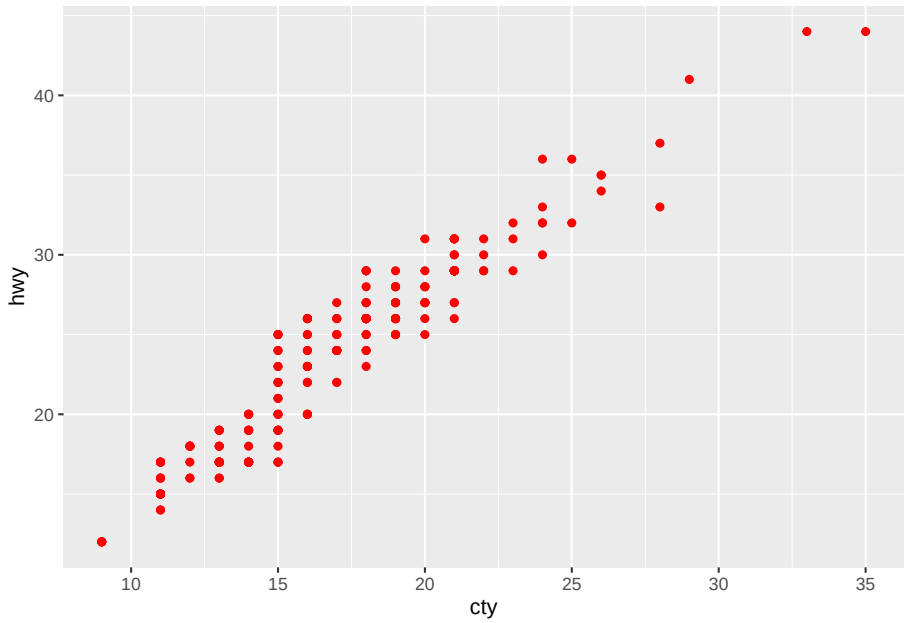

```
ggplot(dados) +
  geom_point(aes(x = cty, y = hwy))
```



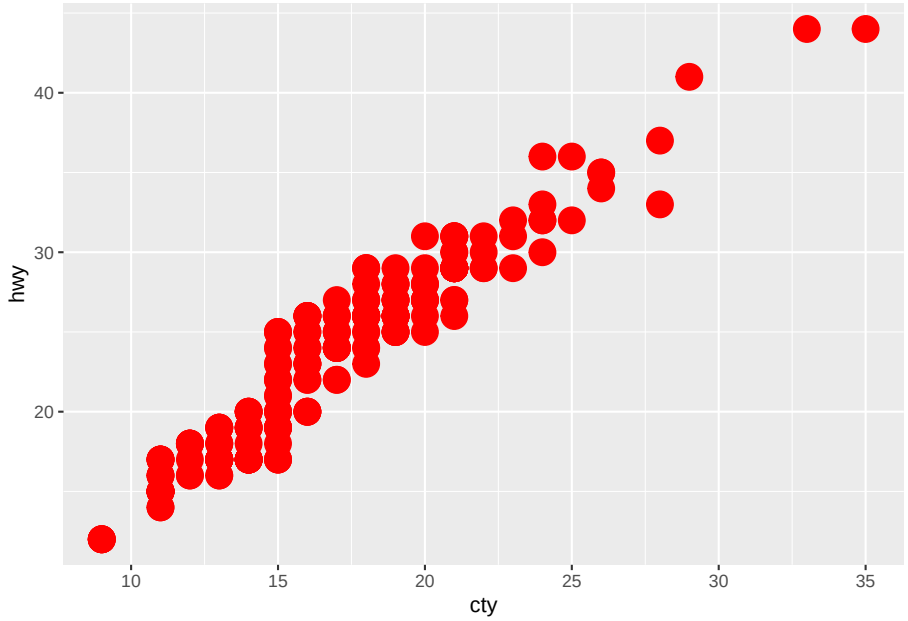

```
ggplot() +
  geom_point(data = dados, aes(x = cty, y = hwy))
```



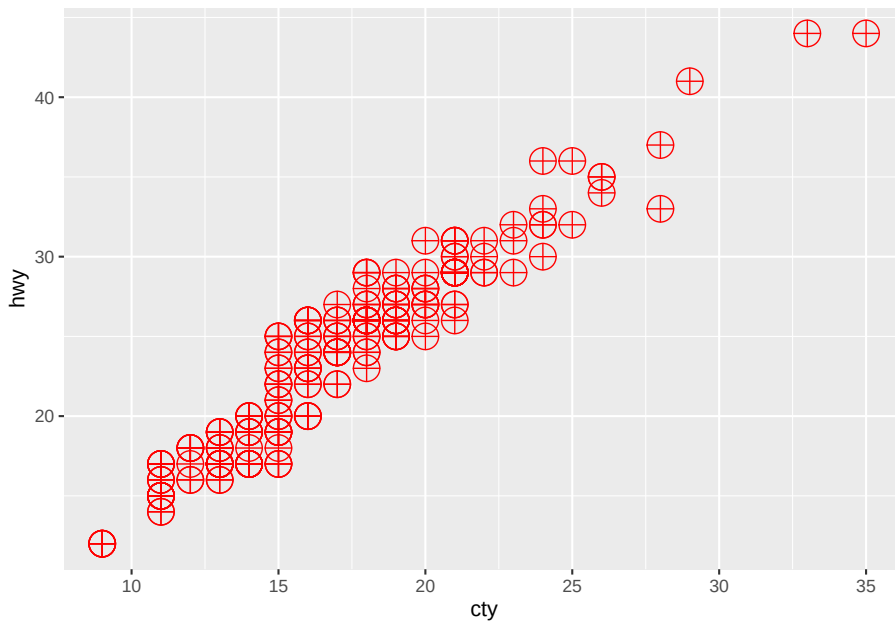

```
# Fim

ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red")
```



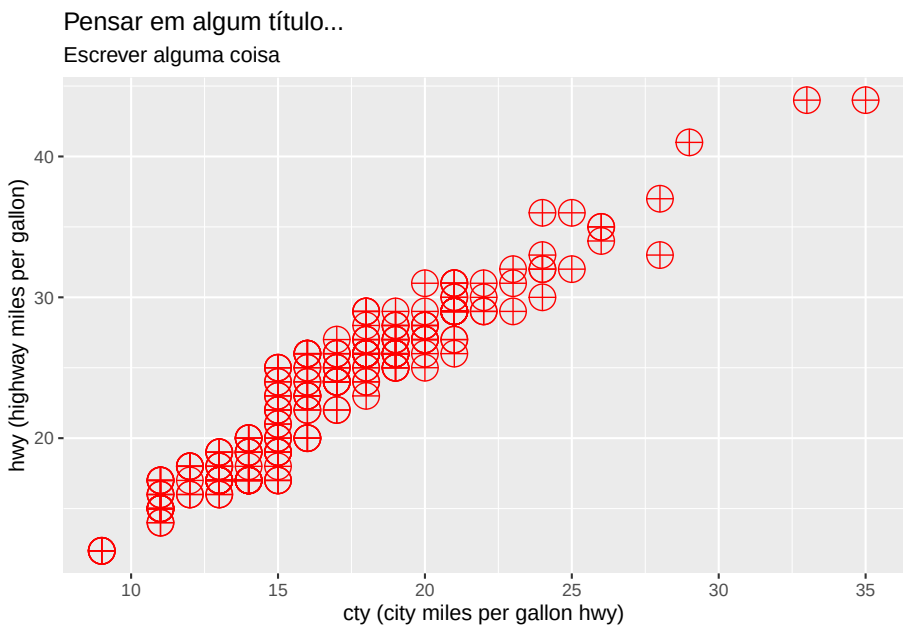

```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red", size = 6)
```




```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red", size = 6, shape = 10)
```




```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red", size = 6, shape = 10)+
  labs(x = "cty (city miles per gallon hwy)",
       y = "hwy (highway miles per gallon)",
       title = "Pensar em algum título...",
       subtitle = "Escrever alguma coisa")
```



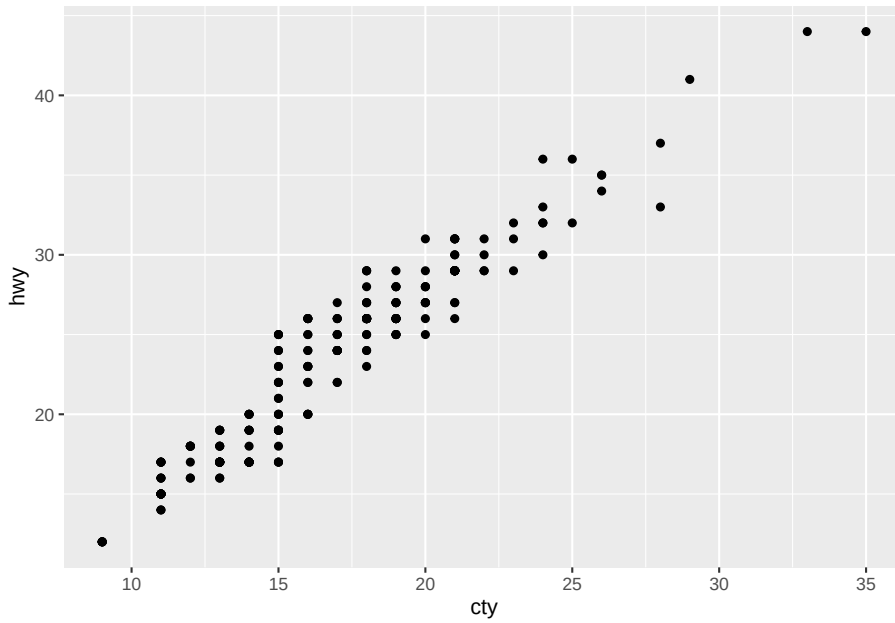
Mais detalhes sobre geom_point

geom_point() understands the following aesthetics (required aesthetics are in bold):

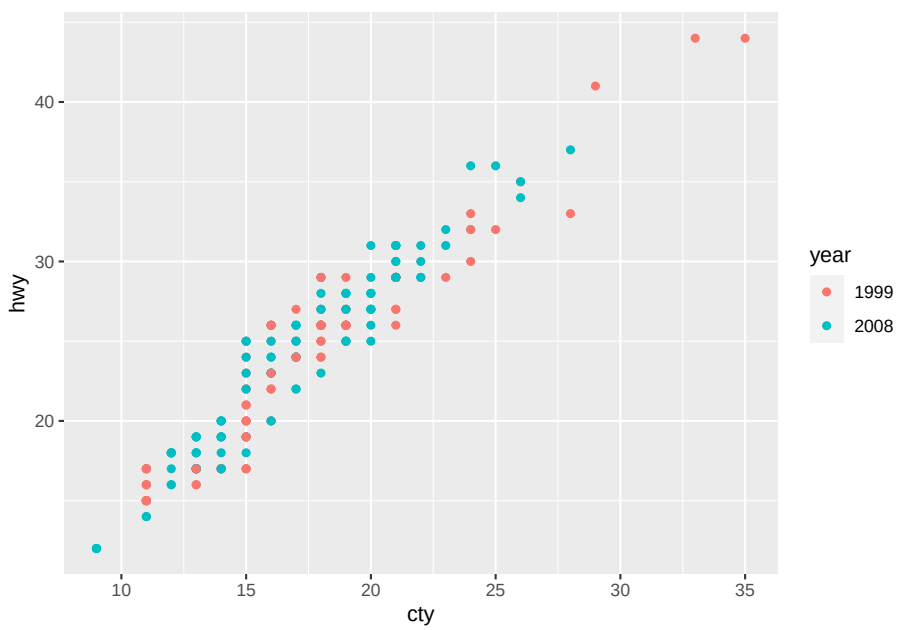
- x
- y
- alpha
- colour
- fill
- group
- shape
- size
- stroke

3.2 colour, shape, fill, alpha

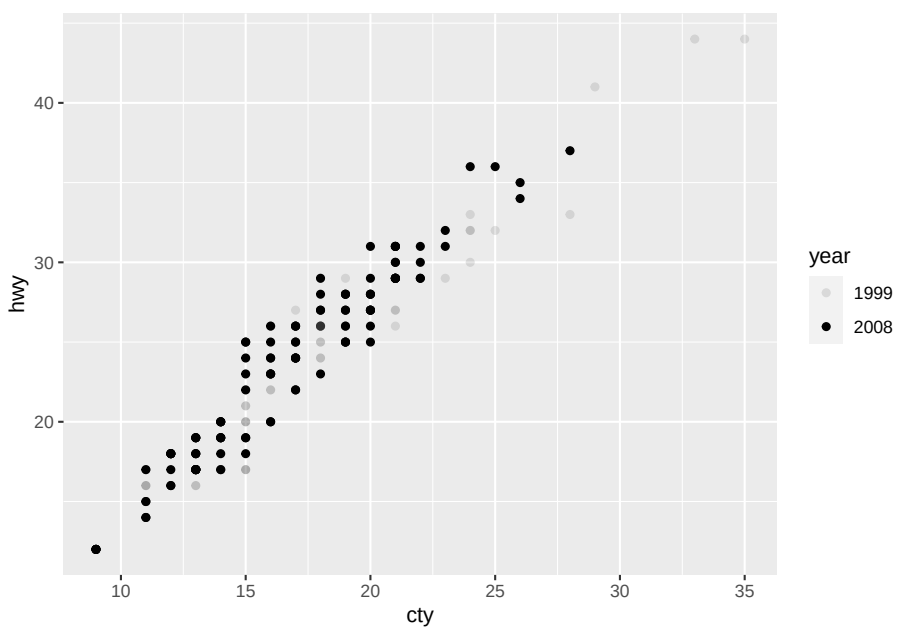
```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point()
```



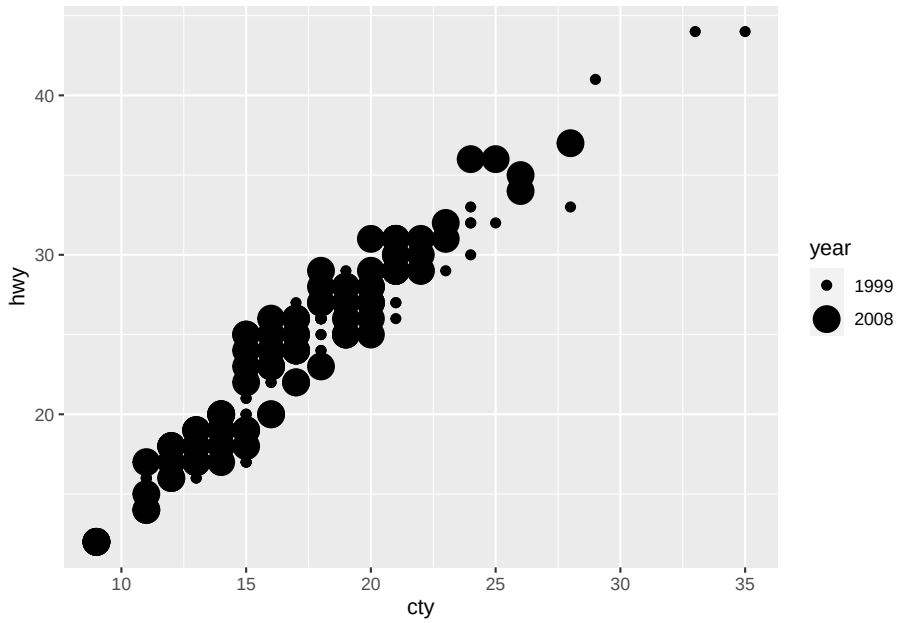

```
ggplot(dados, aes(x = cty, y = hwy, col = factor(year))) +
  geom_point() +
  labs(col = "year")
```



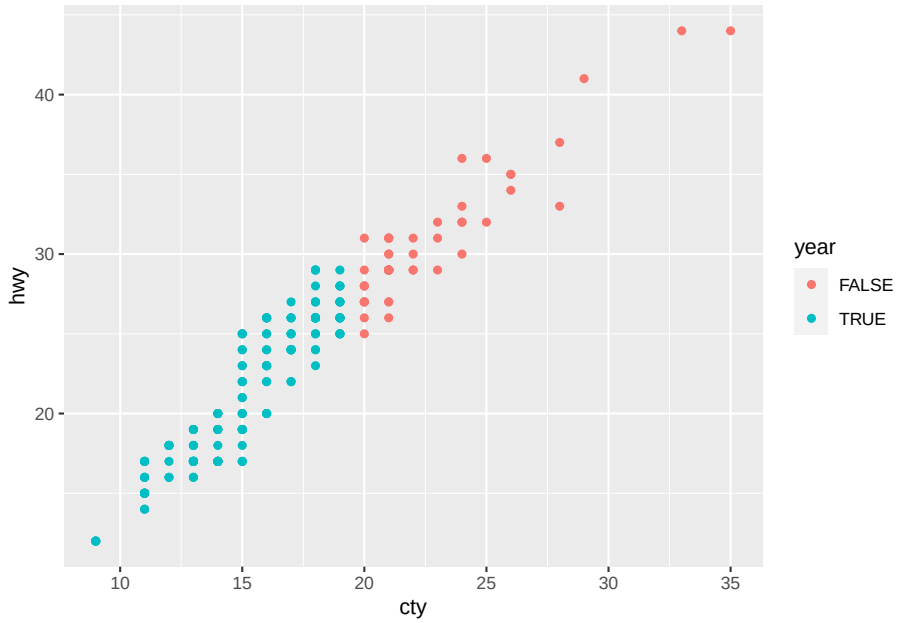
```
ggplot(dados, aes(x = cty, y = hwy, alpha = factor(year))) +
  geom_point() +
  labs(alpha = "year")
```



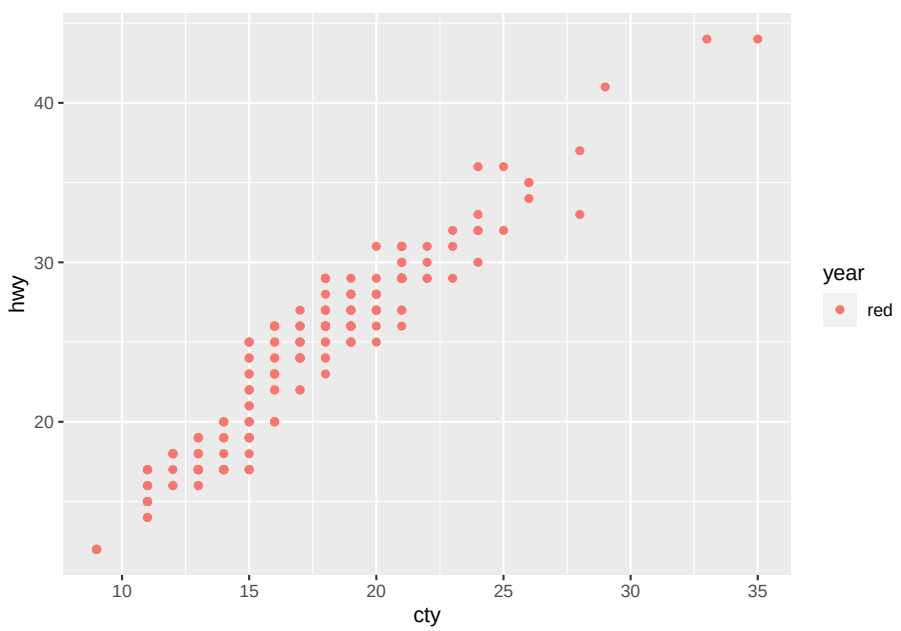

```
ggplot(dados, aes(x = cty, y = hwy, size = factor(year))) +
  geom_point() +
  labs(size = "year")
```



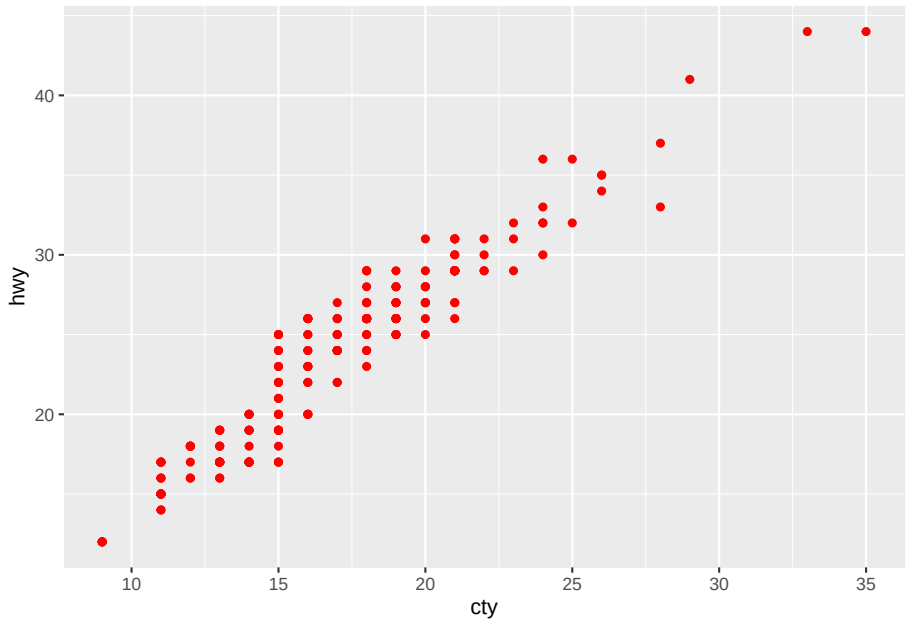

```
# Alternativa
ggplot(dados, aes(x = cty, y = hwy, col = cty < 20)) +
  geom_point() +
  labs(col = "year")
```



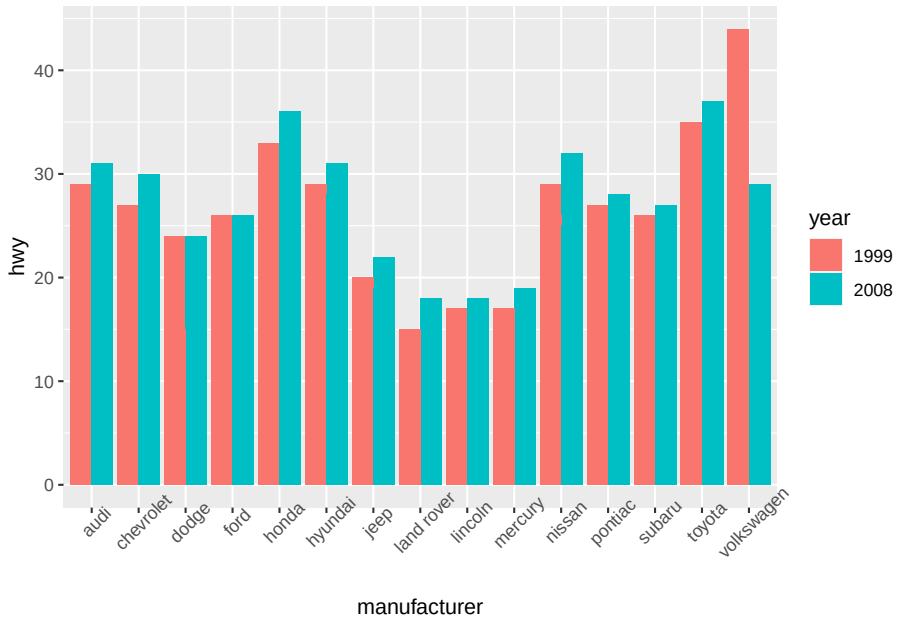
```
# Erro comum
ggplot(dados, aes(x = cty, y = hwy, col = "red")) +
  geom_point()+
  labs(col = "year")
```



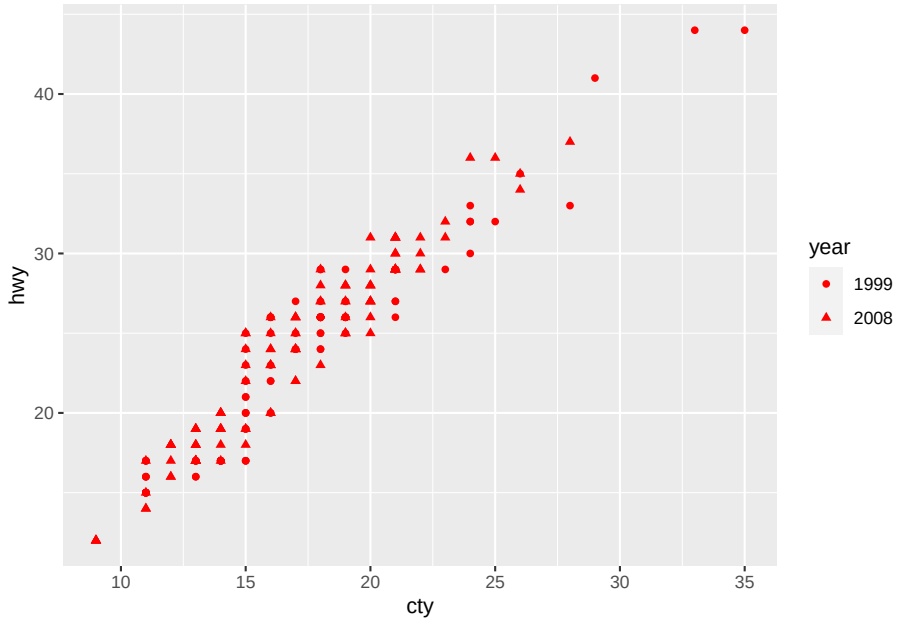

```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(col = "red")+
  labs(col = "year")
```



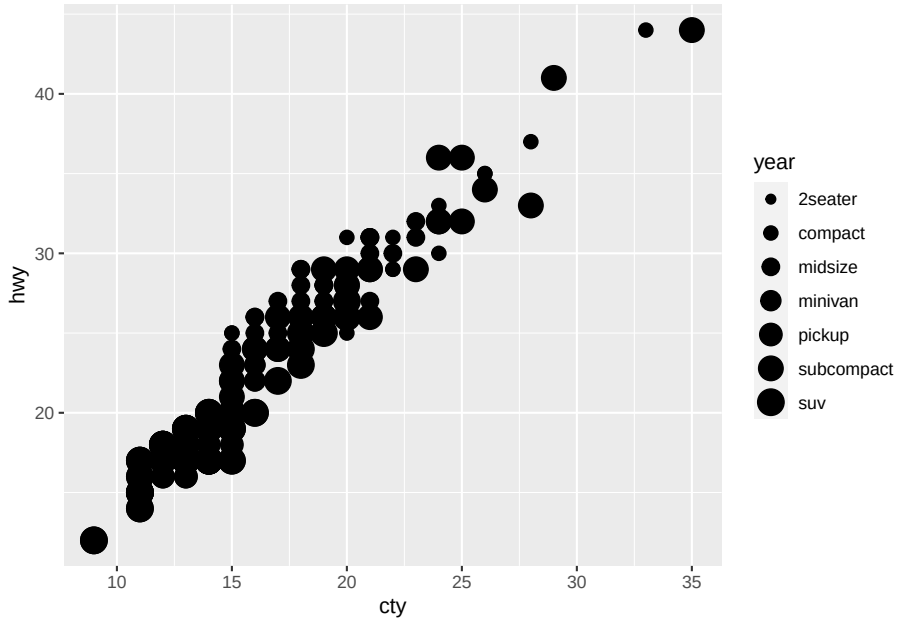

```
# Fim  Erro comum

ggplot(dados, aes(x = manufacturer, y = hwy, fill = factor(year))) +
  geom_col(position = "dodge") +
  labs(fill = "year") +
  theme(axis.text.x = element_text(angle = 45))
```



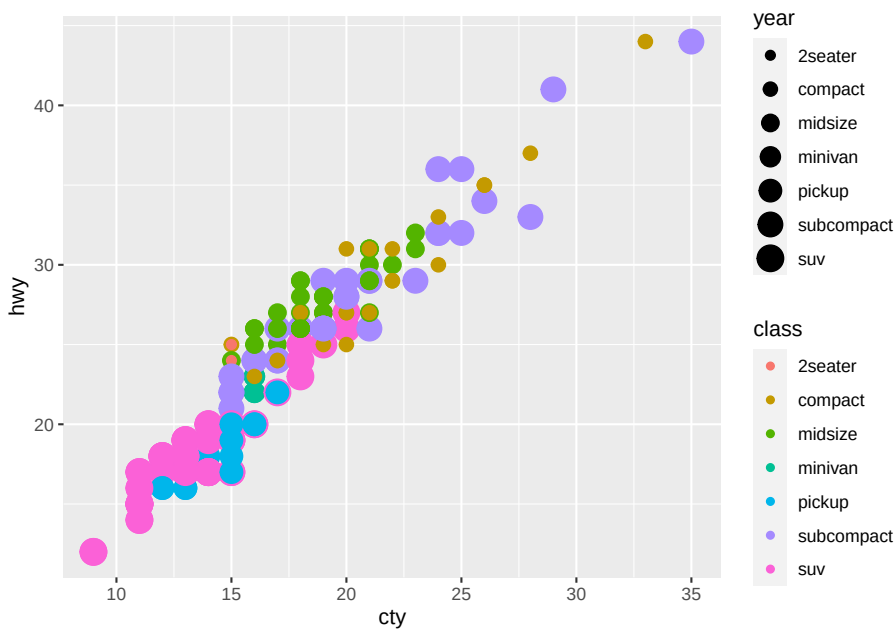

```
ggplot(dados, aes(x = cty, y = hwy, shape = factor(year))) +
  geom_point(col = "red") +
  labs(shape = "year")
```



```
ggplot(dados, aes(x = cty, y = hwy, size = factor(class))) +
  geom_point() +
  labs(size = "year")
```




```
ggplot(dados, aes(x = cty, y = hwy, size = factor(class), col = factor(class))) +
  geom_point() +
  labs(size = "year",
       col = "class")
```

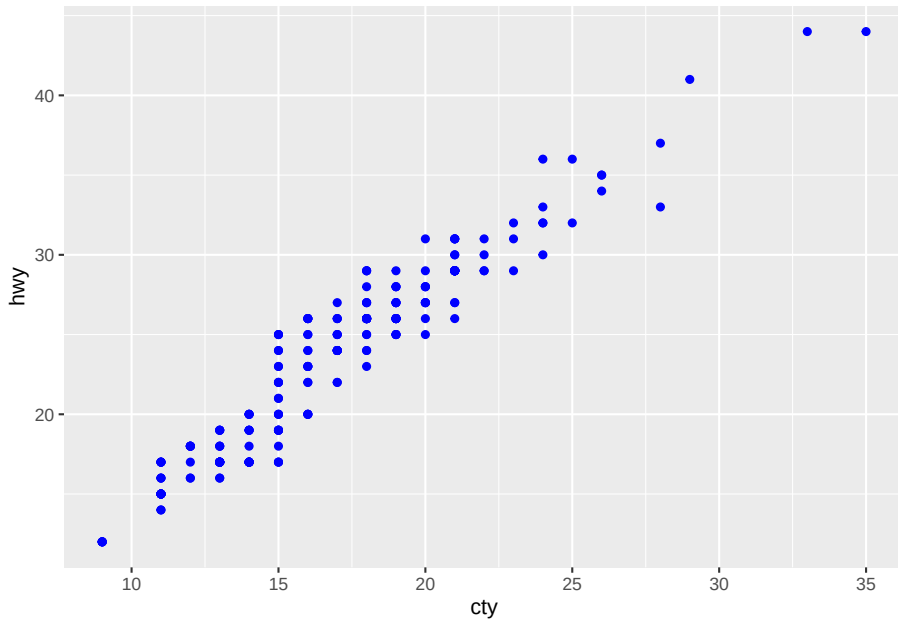


3.3 point, smooth, boxplot, histogram

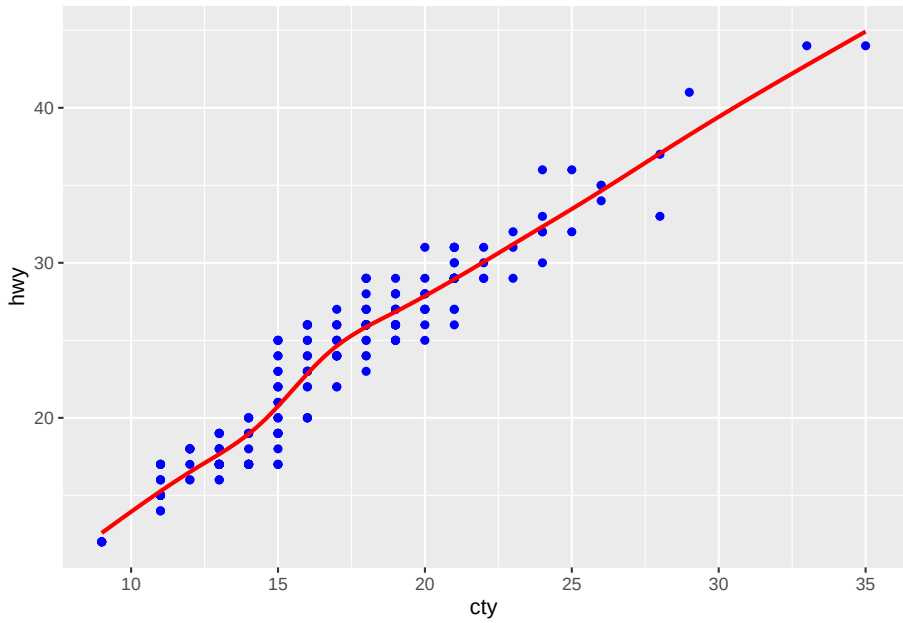
```
glimpse(dados)
```

```
## Rows: 234
## Columns: 14
## $ manufacturer          <chr> "audi", "audi", "audi", "audi", "audi", "audi",~
## $ model                 <chr> "a4", "a4", "a4", "a4", "a4", "a4", "a4", "a4 q~
## $ displ                 <dbl> 1.8, 1.8, 2.0, 2.0, 2.8, 2.8, 3.1, 1.8, 1.8, 2.~
## $ year                  <int> 1999, 1999, 2008, 2008, 1999, 1999, 2008, 1999,~
## $ cyl                   <int> 4, 4, 4, 4, 6, 6, 6, 4, 4, 4, 4, 6, 6, 6, 6, 6,~
## $ trans                 <chr> "auto(l5)", "manual(m5)", "manual(m6)", "auto(a~
## $ drv                   <chr> "f", "f", "f", "f", "f", "f", "f", "4", "4", "4~
## $ cty                   <int> 18, 21, 20, 21, 16, 18, 18, 18, 16, 20, 19, 15,~
## $ hwy                   <int> 29, 29, 31, 30, 26, 26, 27, 26, 25, 28, 27, 25,~
## $ fl                    <chr> "p", "p", "p", "p", "p", "p", "p", "p", "p", "p~
## $ class                 <chr> "compact", "compact", "compact", "compact", "co~
## $ `avg miles per gallon` <dbl> 23.5, 25.0, 25.5, 25.5, 21.0, 22.0, 22.5, 22.0,~
## $ car                   <chr> "audi a4", "audi a4", "audi a4", "audi a4", "au~
## $ `cyl / trans`         <chr> "4 cylinders / auto(l5) transmission", "4 cylin~
```

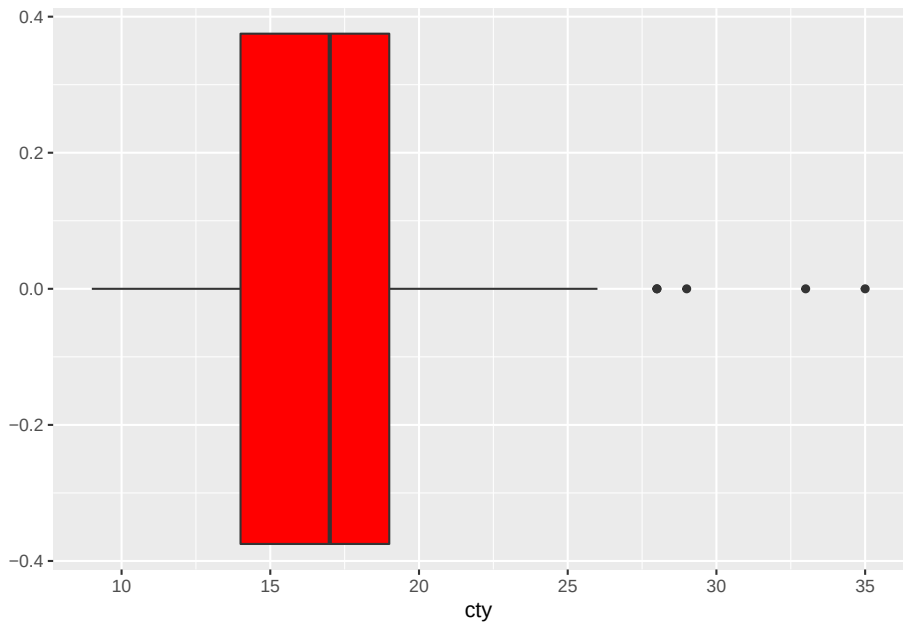
```
v1 <- ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(col = "blue")
v1
```



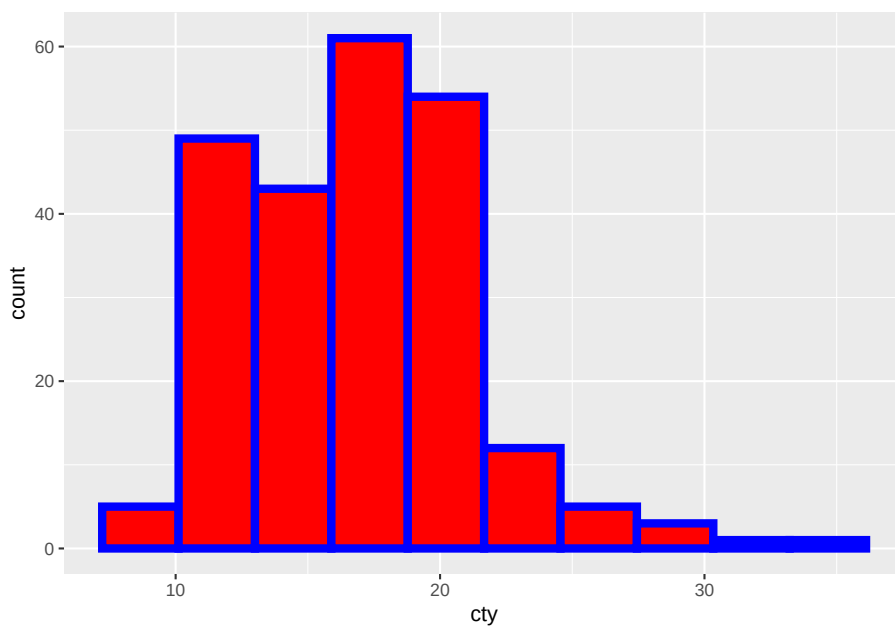

```
v2 <- ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(col = "blue")+
  geom_smooth(method = mgcv::gam,
              formula = y ~ s(x, bs = "cs") ,
              col = "red",
              se = FALSE)
v2
```



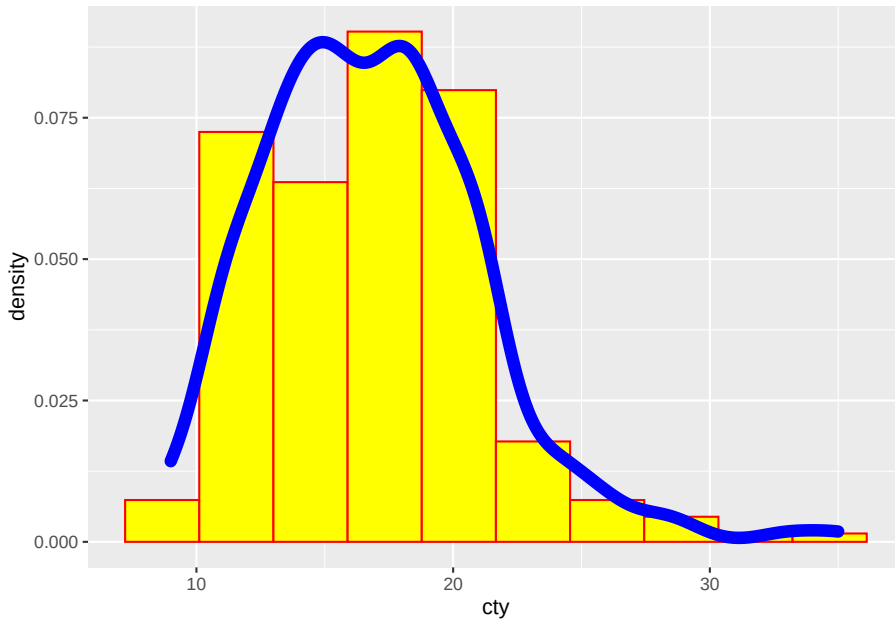

```
v3 <- ggplot(dados, aes(x = cty)) +
  geom_boxplot(fill = "red")
v3
```



```
v4 <- ggplot(dados, aes(x = cty)) +
  geom_histogram(bins = 10, fill = "red", col = "blue", lwd=2)
v4
```




```
# Alternativas
ggplot(dados, aes(x = cty)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 10, fill = "yellow", col = "red") +
  geom_density(col = "blue", lwd =3)
```



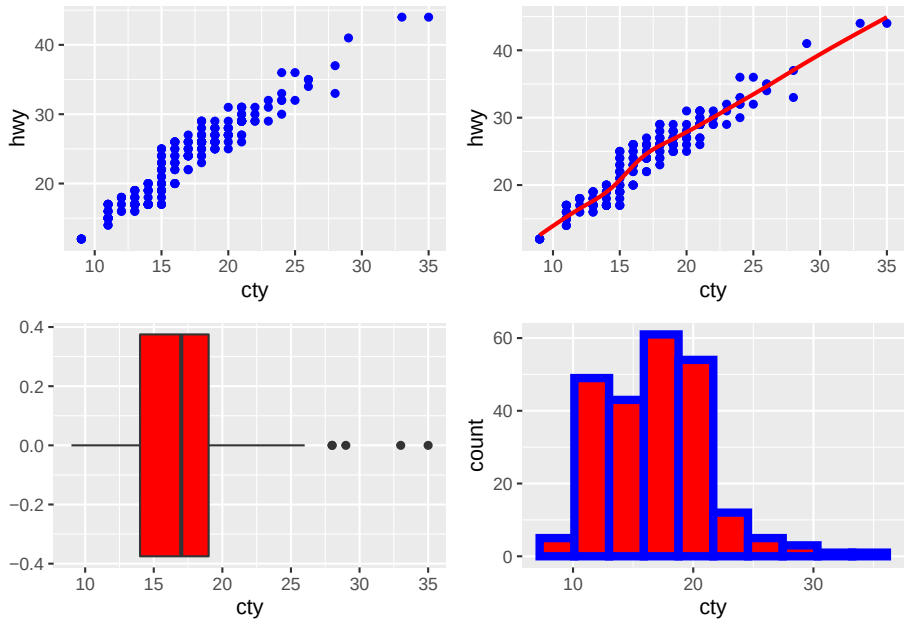
3.4 gridExtra e patchwork

Alguns links

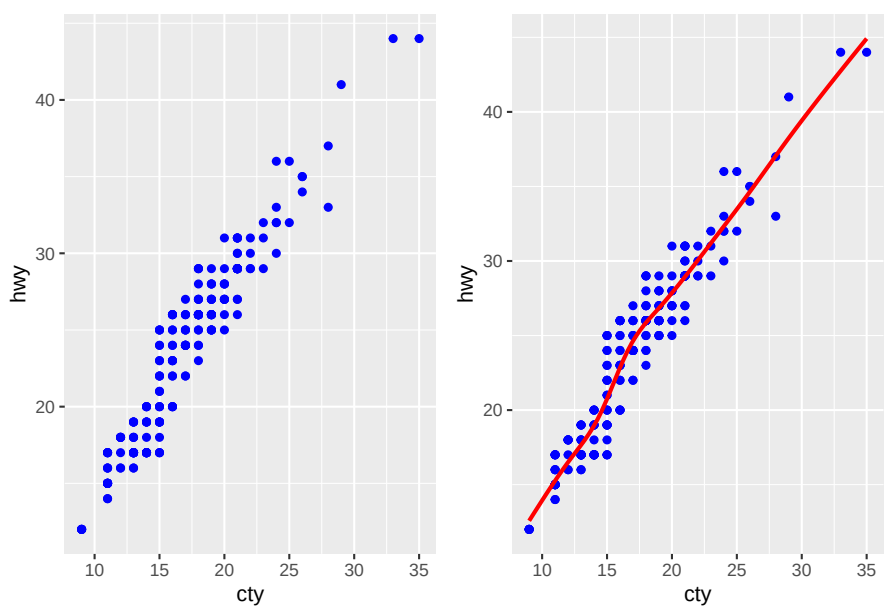
link 1: patchwork

link 2: patchwork

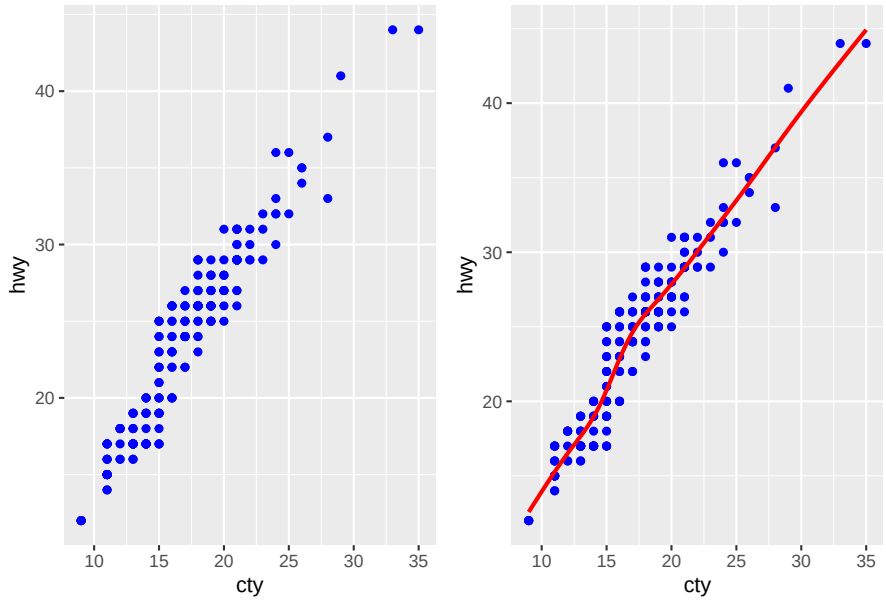
```
# gridExtra
grid.arrange(v1, v2, v3, v4)
```



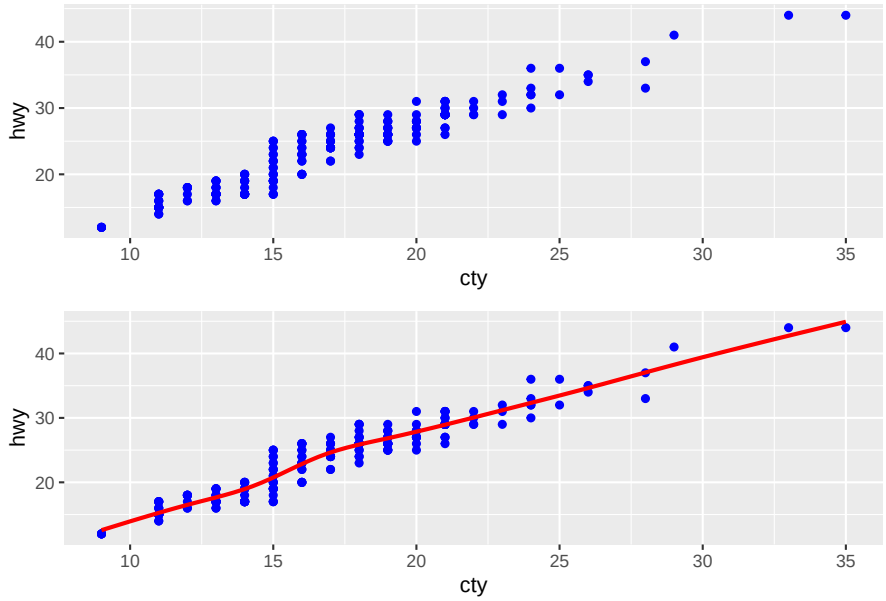

```
# patchwork
v1 + v2
```



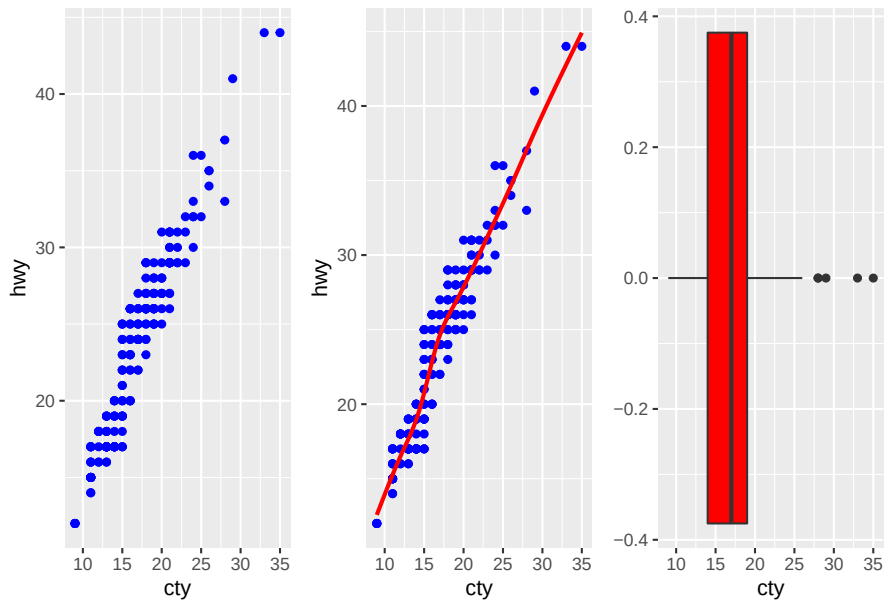
```
v1 | v2
```


hwy
40
30
20
10
15
20
25
30
35
cty
hwy
40
30
20
10
15
20
25
30
35
cty

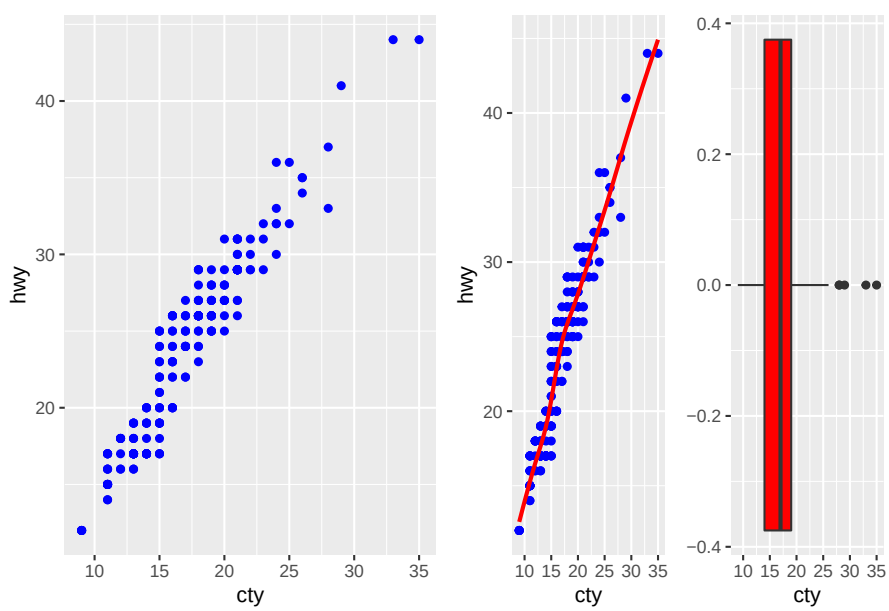
```
v1 / v2
```


hwy
40
30
20
10
15
20
25
30
35
cty
hwy
40
30
20
10
15
20
25
30
35
cty

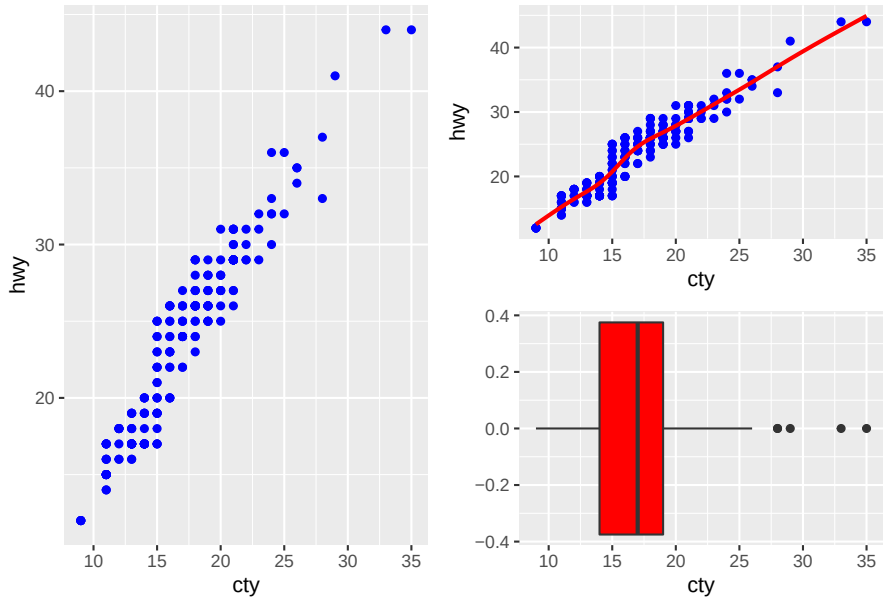
```
v1 + v2 + v3
```



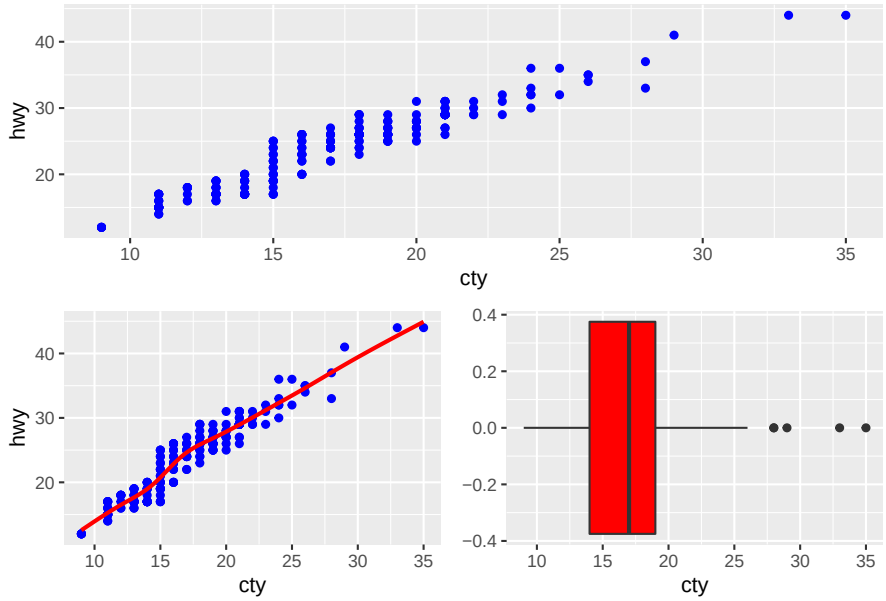

```
v1 + (v2 + v3)
```



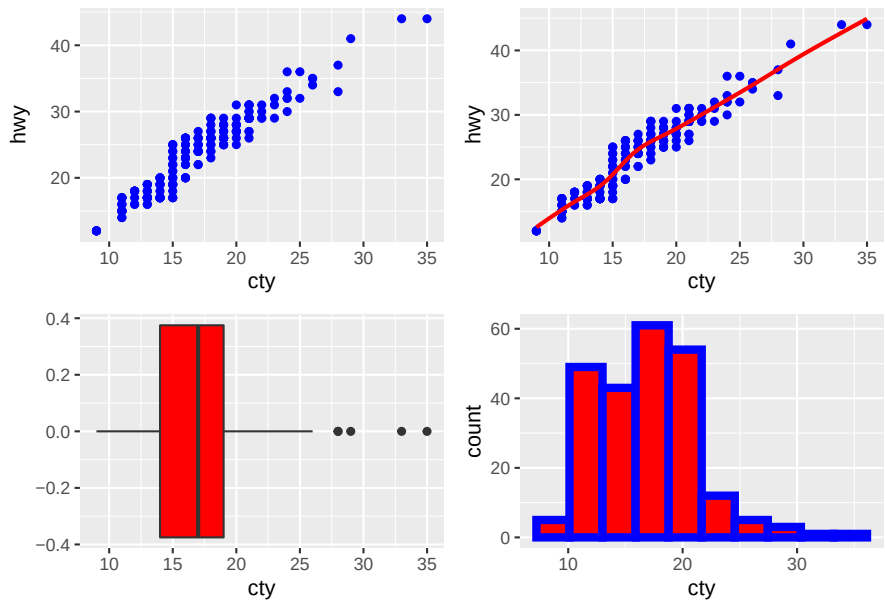
```
v1 | (v2 / v3)
```



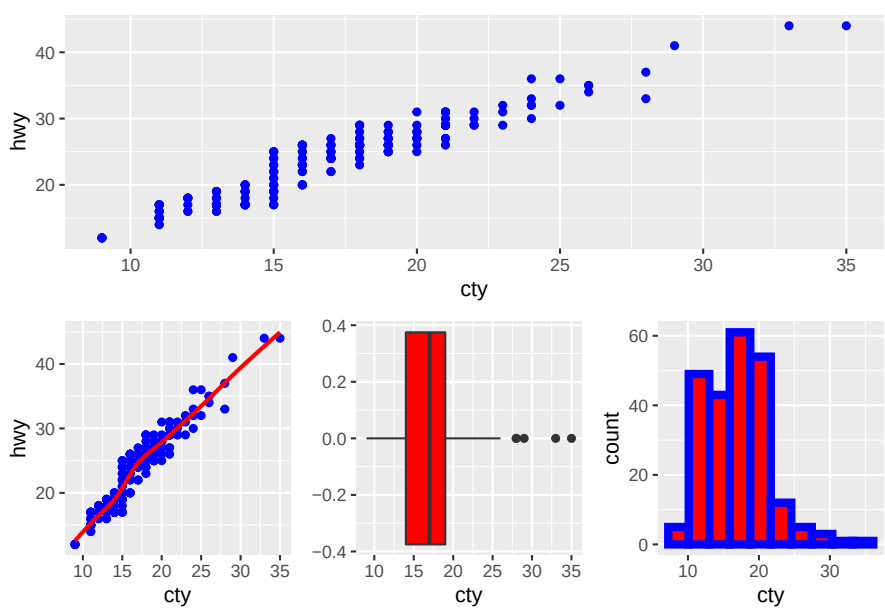

```
v1 / (v2 + v3)
```



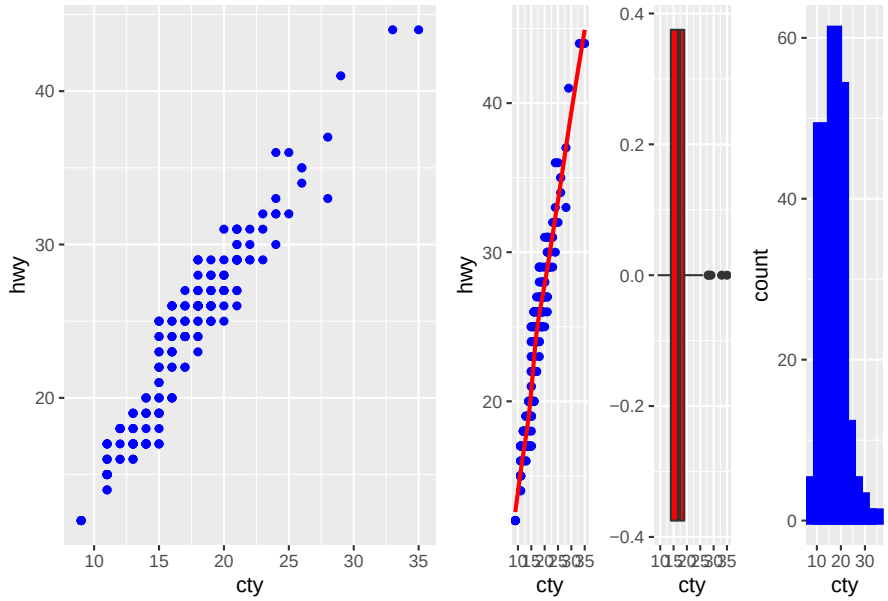
```
v1 + v2 + v3 + v4
```



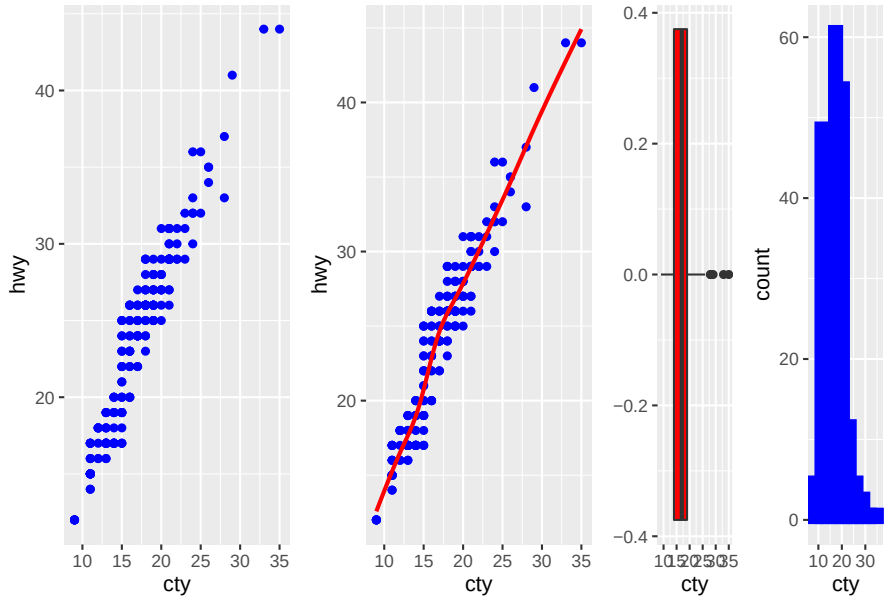

```
v1/(v2+v3+v4)
```



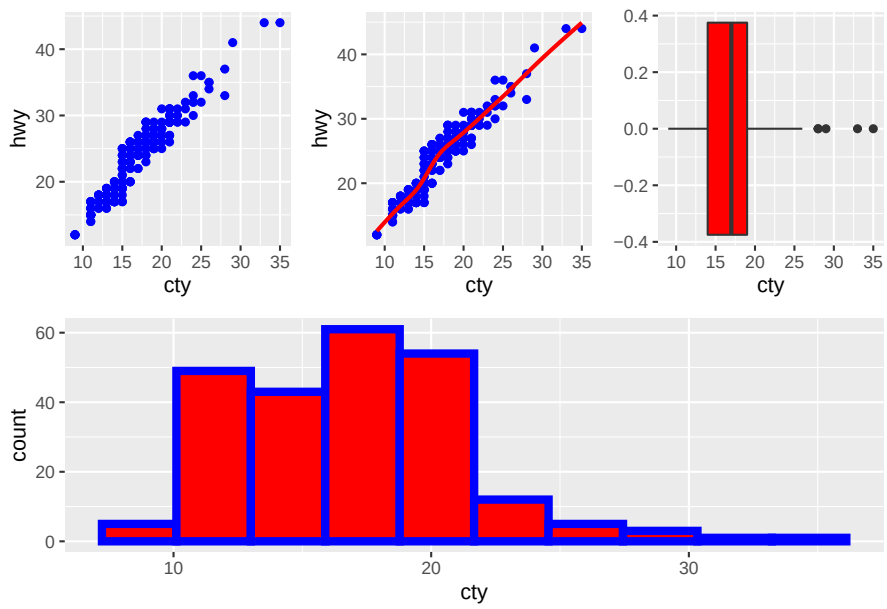
```
v1  + (v2 + v3 + v4)
```



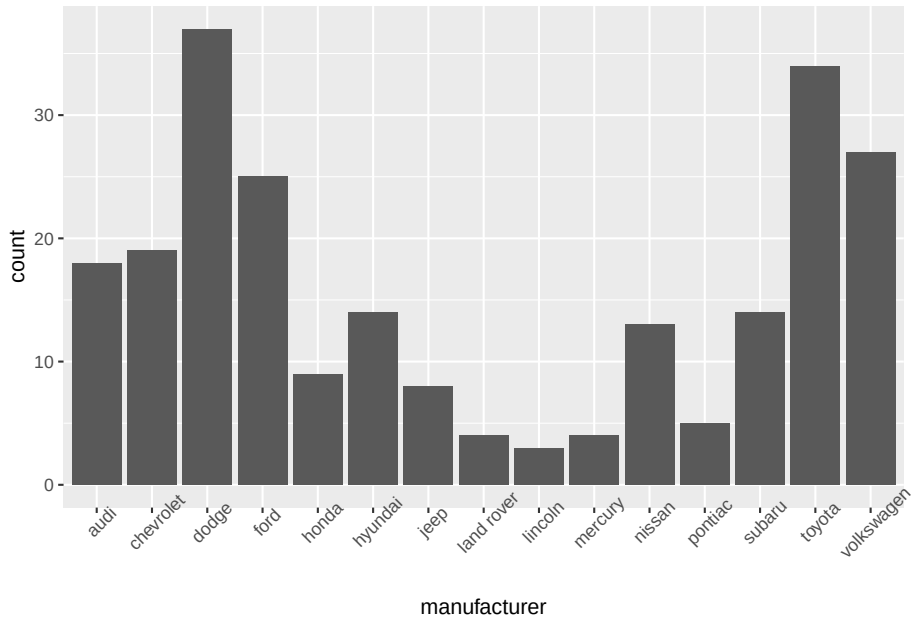

```
v1  + v2 + (v3 + v4)
```



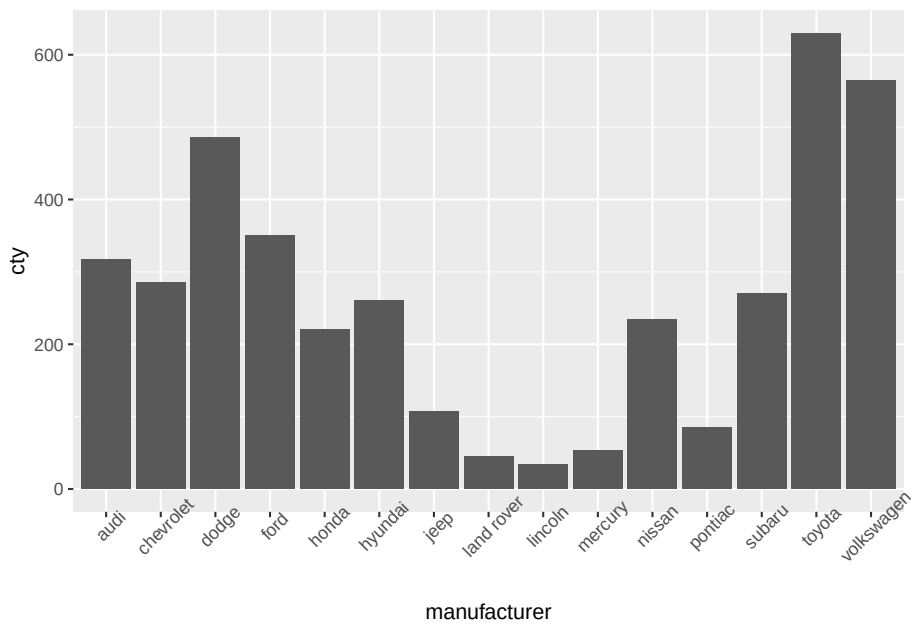
```
(v1 | v2 | v3) / v4
```



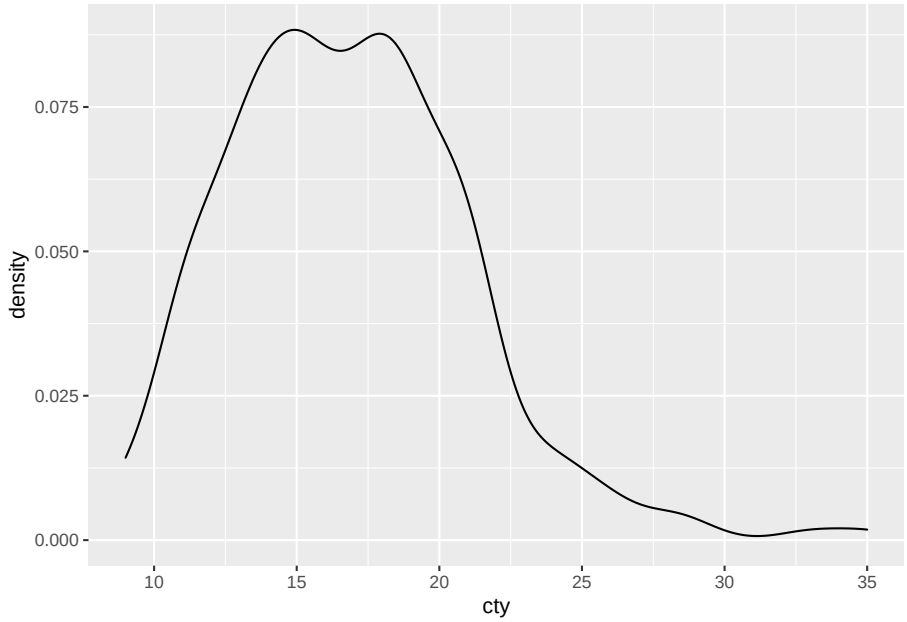

```
v5 <- ggplot(dados , aes(x = manufacturer)) +
  geom_bar()+
  theme(axis.text.x = element_text(angle = 45))
v5
```



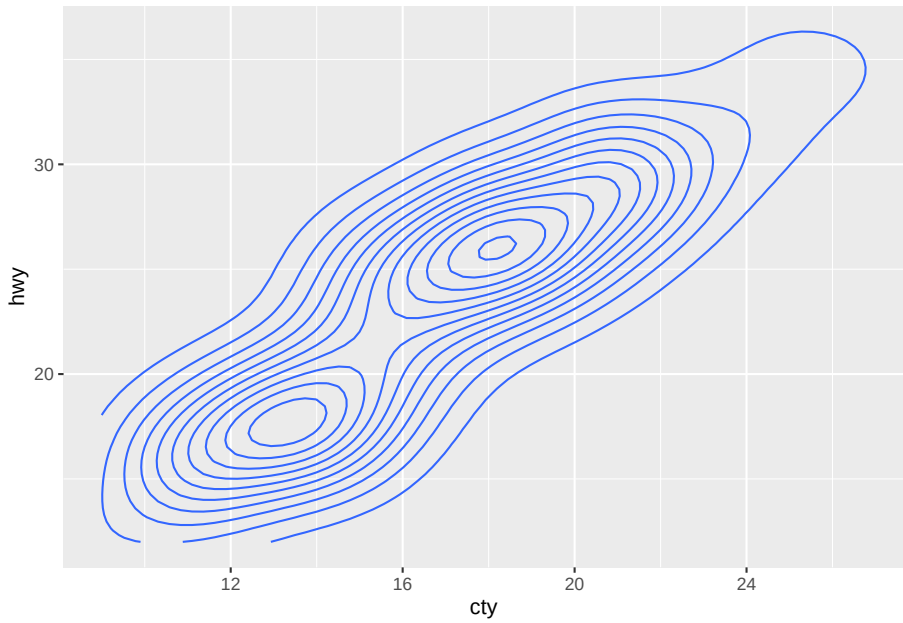

```
v6 <- ggplot(dados , aes(x = manufacturer, y = cty)) +
  geom_col()+
  theme(axis.text.x = element_text(angle = 45))
v6
```



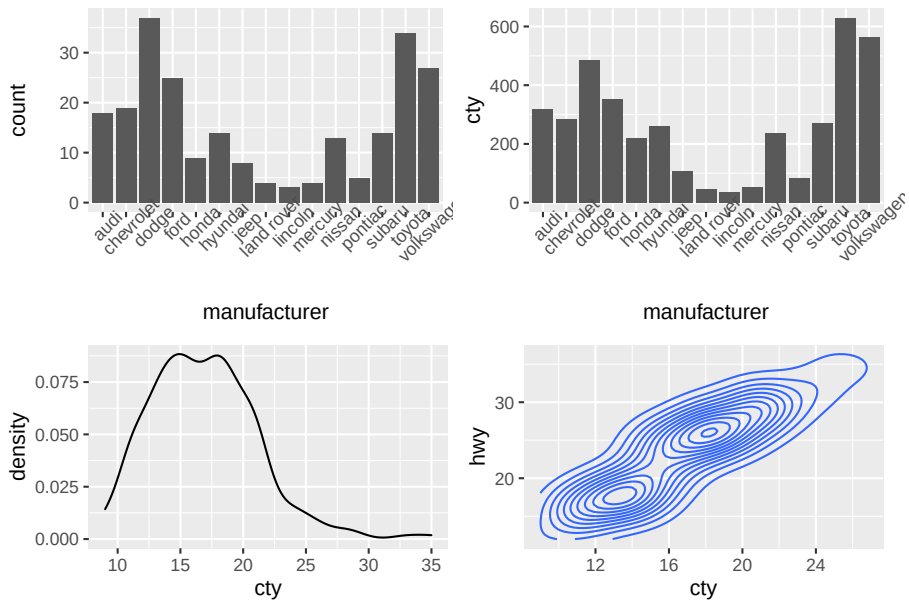
```
v7 <- ggplot(dados , aes(x = cty)) +
  geom_density()
v7
```



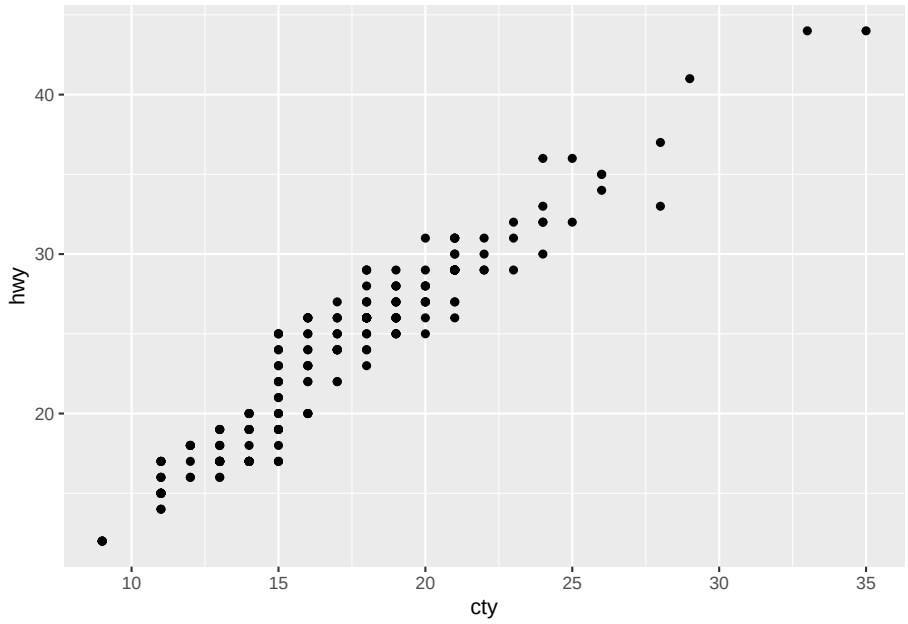

```
v8 <- ggplot(dados, aes(x = cty, y = hwy)) +
  geom_density2d()
v8
```


hwy
30
20
12
16
20
24
cty

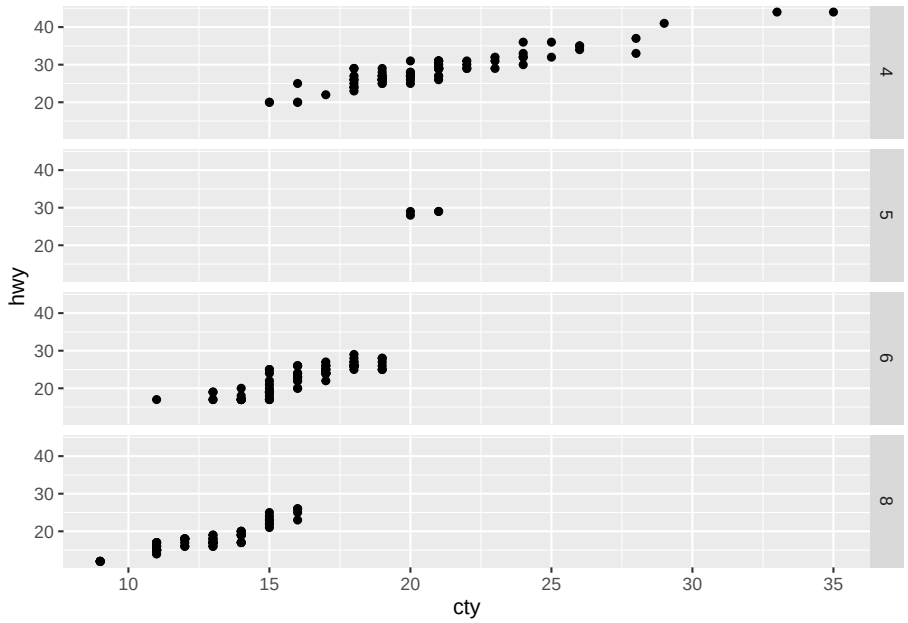
```
(v5+v6)/ (v7 + v8)
```


count
30
20
10
0
audi
chevrolet
dodge
ford
honda
hyundai
jeep
land rover
lincoln
mercury
nissan
pontiac
subaru
toyota
volkswagen
manufacturer
cty
600
400
200
0
audi
chevrolet
dodge
ford
honda
hyundai
jeep
land rover
lincoln
mercury
nissan
pontiac
subaru
toyota
volkswagen
manufacturer
density
0.075
0.050
0.025
0.000
10
15
20
25
30
35
cty
hwy
30
20
12
16
20
24
cty

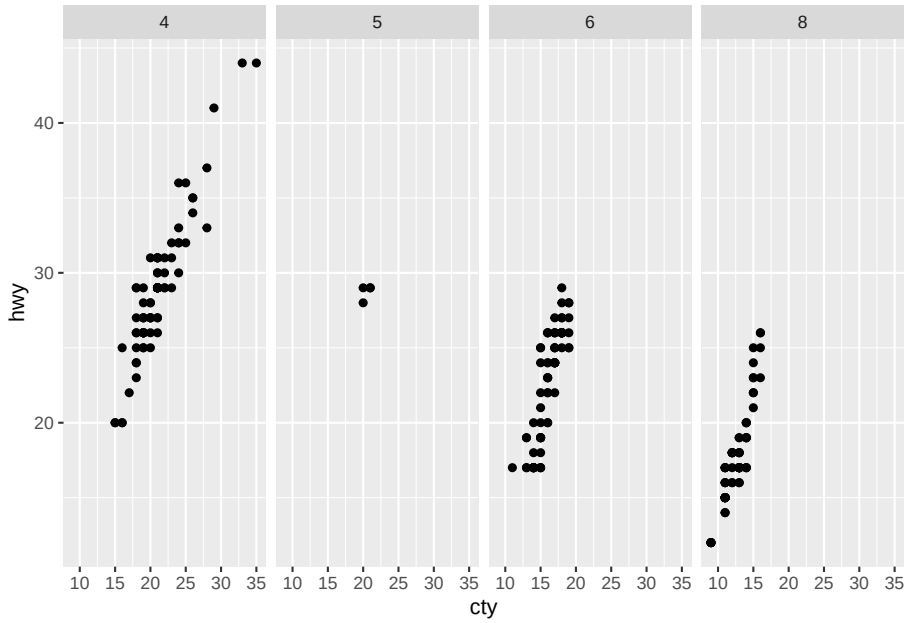
3.6 facet_grid, facet_grid



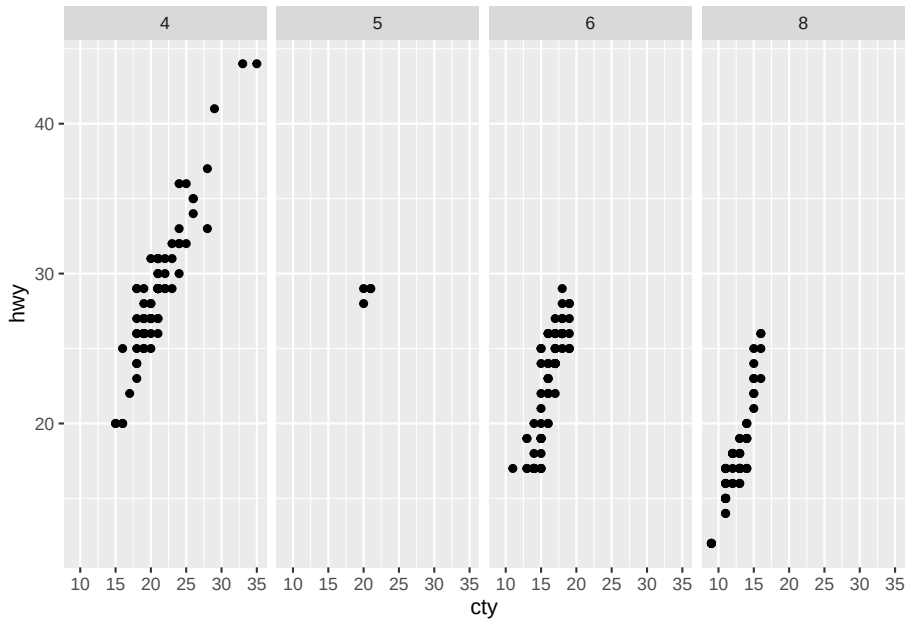

```
p1 + facet_grid(rows = vars(cyl))
```



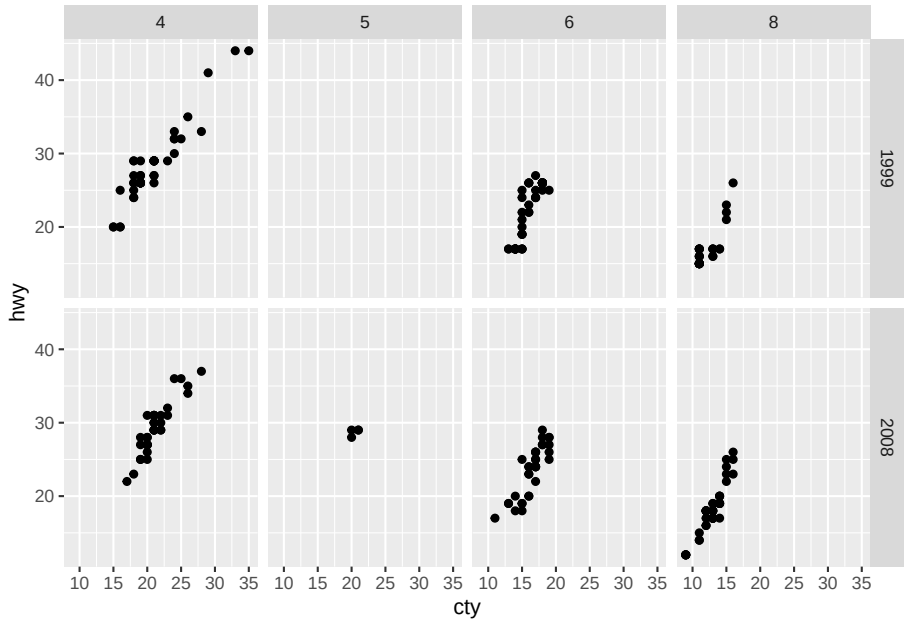

```
p1 + facet_grid(cols = vars(cyl))
```



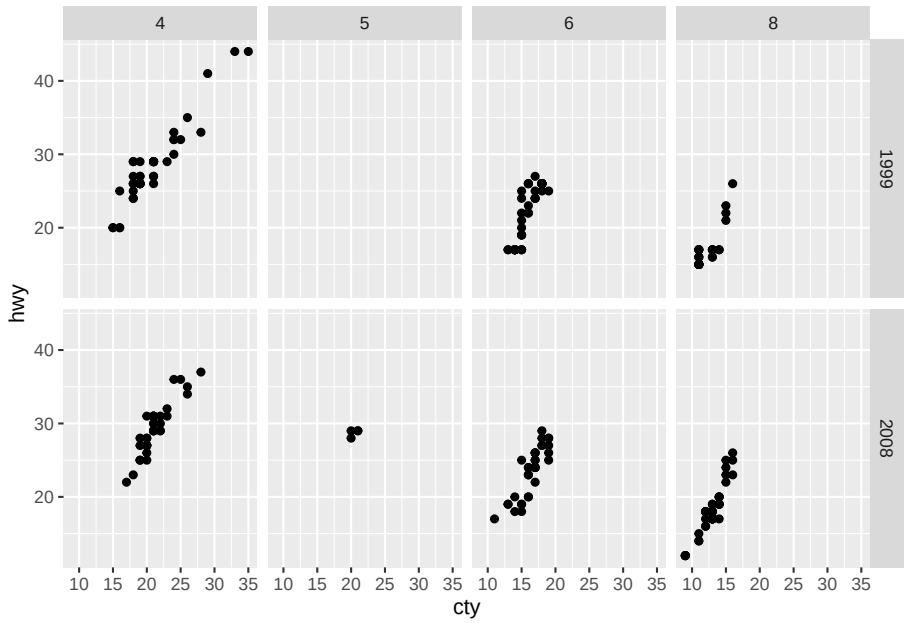

```
p1 + facet_grid(~cyl)
```



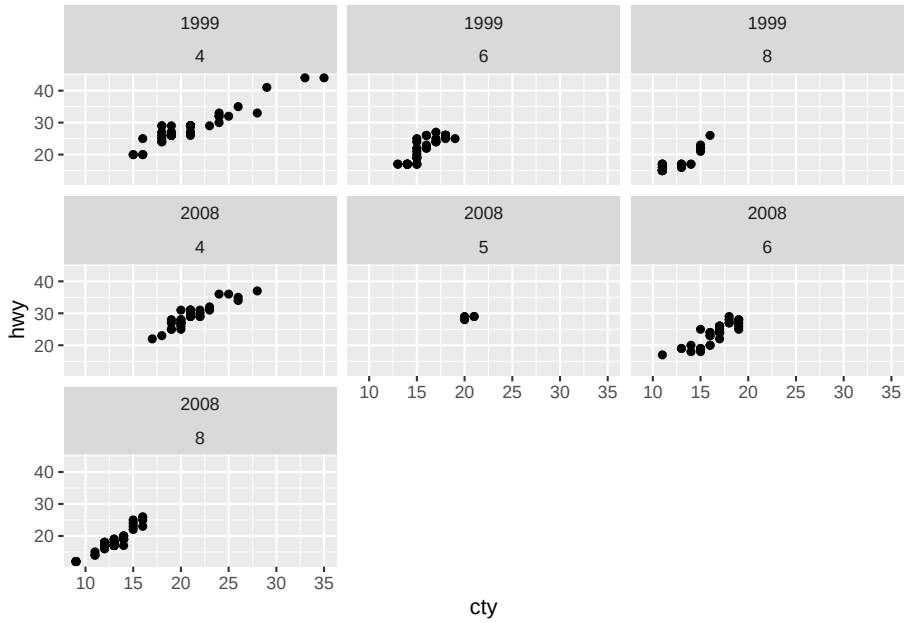

```
p1 + facet_grid(rows = vars(year), cols =vars(cyl))
```



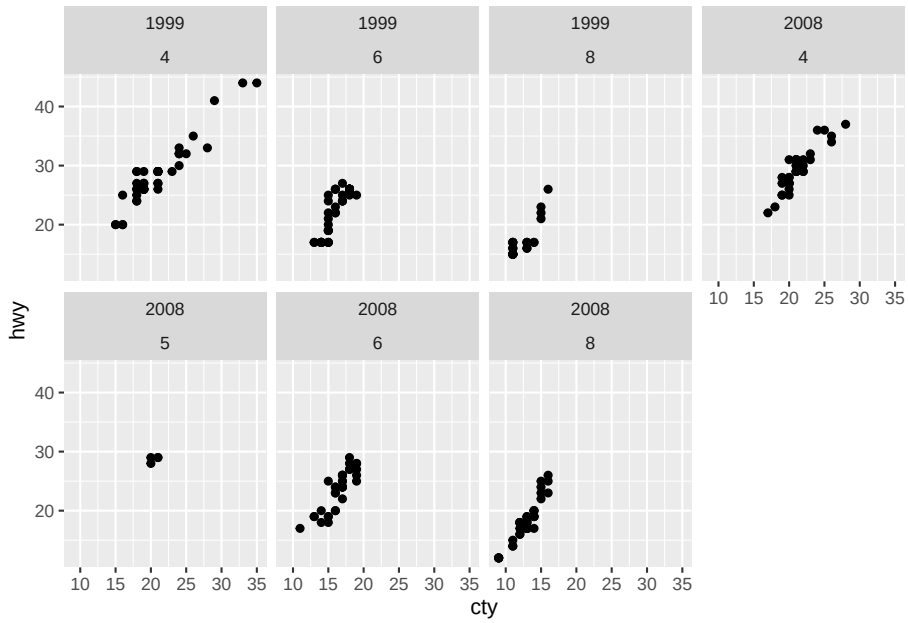

```
p1 + facet_grid(year~cyl)
```



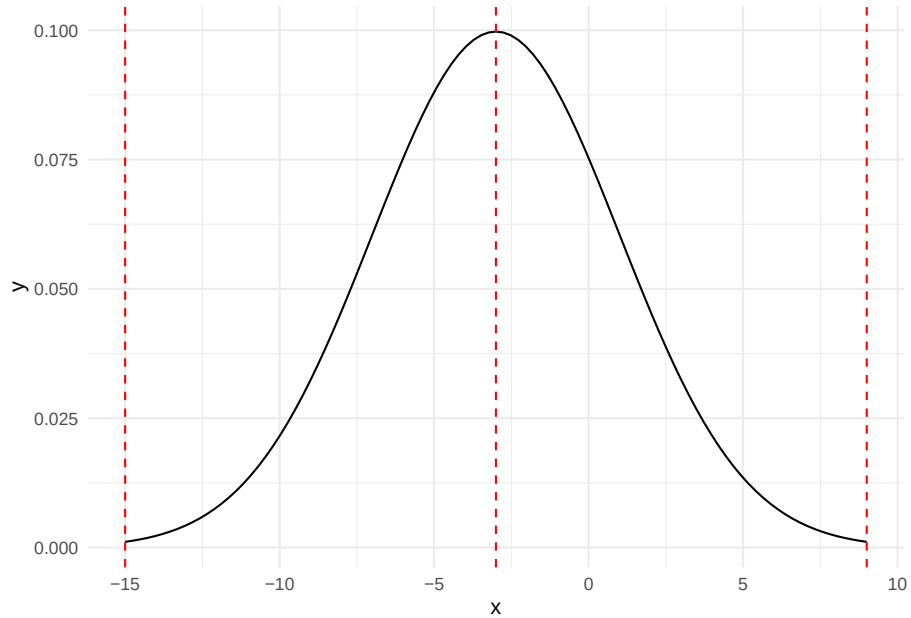

```
p1 + facet_wrap(year~cyl)
```



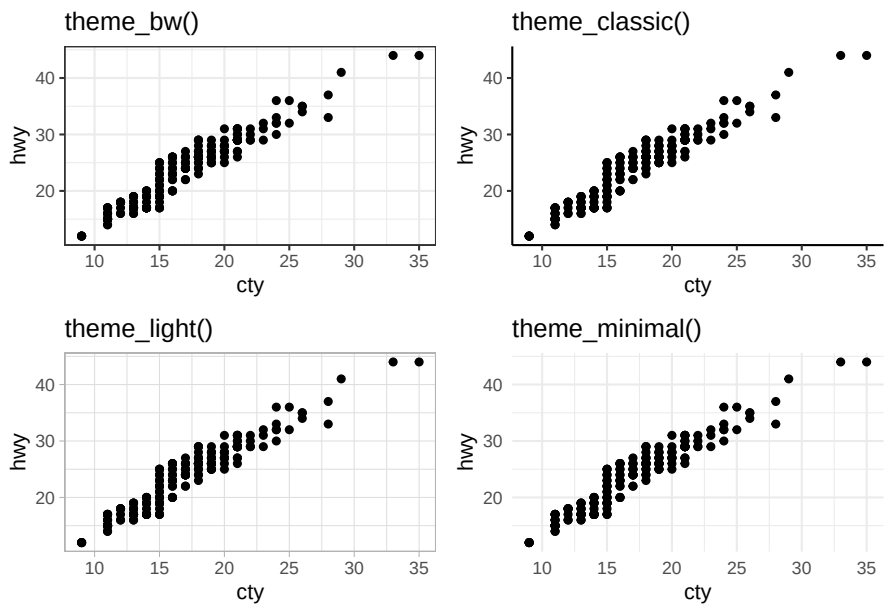

```
p1 + facet_wrap(year~cyl, ncol = 4)
```




```
a<- -3 # média
b<- 4  # desv. padrão
ggplot(data.frame(x = c(a - 3*b, a + 3*b)), aes(x)) +
  stat_function(fun = dnorm, args = list(mean = a, sd = b))+
  geom_vline(xintercept = c(a - 3*b, a, a + 3*b), col = "red", lty = 2)+
  theme_minimal()
```




```
a1<- p1 + theme_bw() + labs(title = "theme_bw()")
a2<- p1 + theme_classic() + labs(title = "theme_classic()")
a3<- p1 + theme_light() + labs(title = "theme_light()")
a4<- p1 + theme_minimal() + labs(title = "theme_minimal()")

a1 + a2 + a3 + a4
```



3.9 plotly

plotly cran

Interactive web-based data visualization with R, plotly, and shiny

Plotly R Open Source Graphing Library

```
ggplotly(v1)
ggplotly(v2)
ggplotly(v4)
ggplotly(v5)
```

3.10 esquisse

esquisse

esquisse + shiny

Chapter 4

Referências (on-line)

4.1 ggplot2 e dplyr

ggplot2: elegant graphics for data analysis, Hadley Wickham

R Programming for Data Science, Roger D. Peng

R for Data Science, Hadley Wickham e Garrett Grolemund.

R Graphics Cookbook, Winston Chang

4.2 link do dplyr, ggplot2 e magrittr

dplyr do tidyverse

ggplot2 do tidyverse

magrittr do tidyverse?